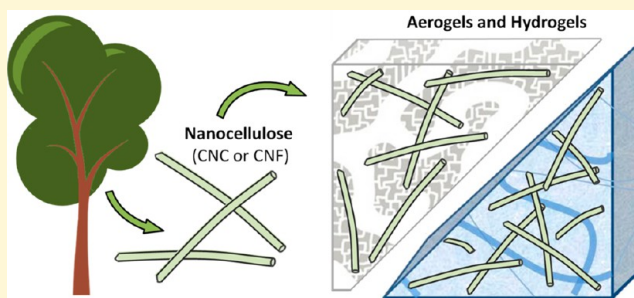


Review of Hydrogels and Aerogels Containing Nanocellulose

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ABSTRACT: Naturally derived cellulose nanocrystals (CNCs) and cellulose nanofibrils (CNFs) are emerging nanomaterials that display high strength, high surface area, and tunable surface chemistry, allowing for controlled interactions with polymers, nanoparticles, small molecules, and biological materials. Industrial production of nanocelluloses is increasing rapidly with several companies already producing on the tons-per-day scale, intensifying the quest for viable products across many sectors. While the hydrophilicity of the nanocellulose interface has posed a challenge to the use of CNCs and CNFs as reinforcing agents in conventional plastics, it is a significant benefit for creating reinforced or structured hydrogel composites (or, when dried, aerogels) exhibiting both mechanical reinforcement and a host of other desirable properties. In this context, this Review describes the quickly growing field of hydrogels and aerogels incorporating nanocelluloses; over 200 references are summarized in comprehensive tables covering the chemistry, preparation, properties, and applications of “nanocellulose-only” and “nanocellulose-containing” gels. Physical and chemical cross-linking strategies, postmodification steps, and routes to control gel structure are discussed, along with key developments and ongoing challenges in the field. Nanocellulose hydrogels and aerogels show great promise in a wide range of biomedical, energy storage, construction, separations, cosmetic, and food applications.



1. INTRODUCTION

Nanocelluloses (cellulose nanocrystals, CNCs, or cellulose nanofibrils, CNFs) are a class of natural, sustainable materials derived from the most abundant renewable polymer on earth. Both CNCs and CNFs have proven extremely versatile, with potential applications spanning the coatings, biomedical, energy, construction, separations, and specialty chemical industries.^{1–3} These high aspect ratio nanoparticles are composed of β (1–4) linked D-glucose units, are partially to fully crystalline, and exhibit impressive mechanical properties and tunable surface chemistries. The physical and chemical properties of nanocelluloses have enabled their use in a wide variety of hydrophilic and hydrophobic composite matrices and hybrid materials, including hydrogels and aerogels. Furthermore, their high surface area-to-volume ratio enables enhanced interactions with, and binding to, polymers, other nanoparticles, and small molecules.⁴ Several nanocellulose reviews have recently been published covering the impressive material properties, new production and processing strategies, characterization, routes for chemical surface modification, biocompatibility/toxicity, and potential applications of CNCs and CNFs, to which the reader is directed for additional information.^{2–14}

CNCs, sometimes termed nanocrystalline cellulose (NCC) or cellulose nanowhiskers (CNW), are rigid rod-shaped particles that are shorter than CNFs (length ca. 100–200 nm, cross-section ca. 5–20 nm). CNCs are generally extracted from natural cellulose sources such as wood pulp or cotton through a sulfuric acid hydrolysis process which leaves negatively charged sulfate half-ester groups on the surface.^{15,16}

This hydrolysis process preferentially cleaves the cellulose

fibers in the amorphous regions, leaving highly crystalline segments intact. Other acids or oxidizing agents can be used to produce CNCs with surface carboxyl groups^{17–19} or phosphate half-esters;²⁰ alternately, HCl hydrolysis leads to uncharged (and colloiddally unstable) CNCs.²¹ Treatment of cellulose with 2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPO) can also impart carboxyl groups on the CNC surface pre- or postextraction.²² Changing CNC suspension conditions such that the surface groups are screened (i.e., through salt addition) or modifying the surface via cationization, silylation, polymer grafting, etc. decreases electrostatic repulsion between particles, leading to reduced colloidal stability and increasing the propensity for gel formation. In addition, uniaxial or chiral nematic alignment of CNCs is possible through the application of shear or external electromagnetic fields and/or cooperative self-assembly processes.^{23–25} Typically, CNCs self-assemble into chiral nematic liquid crystals above a critical concentration of ≈ 4.5 wt %²⁶ but on their own do not gel until concentrations over 10 wt %.²⁷

Conversely, the more flexible and longer CNFs (length >1 μ m, cross-section ca. 5 nm), sometimes termed nanofibrillated cellulose (NFC) or microfibrillated cellulose (MFC), are produced through high energy mechanical homogenization of wood pulp. The delamination/fibrillation process is often combined with enzymatic treatment,²⁸ TEMPO oxidation,²² or other chemical modifications (e.g., carboxymethylation²⁹). As such, CNFs possess a wide variety of surface chemistries and

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Table 1. Properties for “CNC-Only” and Physically Entrapped CNC Hydrogels^a

Materials ^b	CNC Surface Chemistry	wt% CNC (vs. Polymer)	wt% CNC (Total in Gel)	Hydrogel Preparation Method	Storage Modulus (G', kPa)	Compressive Modulus (E, kPa)	Reinforcement	Swelling (pH 7 in H ₂ O)	Intended Application	Ref
“CNC-Only” Hydrogels										
CNC	Sulfate half-ester	100	8.8 – 10.6	Evaporative concentration	≈ 2	–	–	–	–	40
CNC	COOH or NH ₂	100	2.7	pH change	290 or 21	–	–	–	–	41
CNC	Sulfate half-ester	100	2.5 – 5.2	Evaporative concentration	≈ 0.1 – 1	–	–	–	–	81
HEC, HPG, or LBG adsorbed to CNC	Sulfate half-ester/COOH	≈ 94 – 100	0.3 – 19.6	Homogenization	0.001 – 0.1	–	–	–	–	27
CNC	Sulfate half-ester	100	1.5 – 4	Salt-bridging	0.6 – 13.9	–	–	–	–	42
CNC	Sulfate half-ester	100	1 – 4	Hydrothermal heat treatment	≈ 0.1 – 1.5	–	–	–	–	45
Physical Entrapment of CNCs into Hydrated Polymer Networks										
PVA, CNC	Sulfate half-ester	0 – 20	0 – 3	Cyclic freeze-thaw	–	–	–	1.08 – 1.23 x wet	–	48
PVA, CNC	Sulfate half-ester	0 – 7	0 – 0.7	Freeze-thaw	–	800 – 1100	1.38 x	2.63 – 3.37 x dry	Wound dressing	47
PVA, Borax, CNC	Sulfate half-ester	–	1	Homogenization	0.12 – 0.65	0.9 – 4.8	5.3 x	–	Biomedical	49
PVA-MV, CNC-g-DMAEMA, CB[8]	Polymer-grafted	1 – 3	–	Homogenization	3.8 – 14.3	–	–	–	Self-healing	50
PAM, MC, CNC	Sulfate half-ester	–	–	Free radical polymerization, soaked in CNCs	–	≈ 4.5	–	31.8 x dry	–	51
PHEMA, CNC	Sulfate half-ester	–	0.1 – 1	Free radical polymerization	16 – 40	–	–	0.8 – 1.28 x dry	Biomedical	55
CNC, PAM-co-Ac	HCl hydrolyzed	0 – 20	0 – 3.6	Free radical polymerization, oven dried	–	–	–	645 – 1132 x dry	–	58
CNC, starch-g-PSA	HCl hydrolyzed	0 – 20	–	Free radical polymerization, oven dried	–	≈ 11 – 40	≈ 1.2 x	305 – 481 x dry	–	59
CNC, poly N-vinyl caprolactam	Sulfate half-ester	0.2 – 2	–	Homogenization, frontal polymerization	≈ 5 – 20	–	≈ 4 x	≈ 7 – 10 x dry	Biomedical	82
PAM, CNC	Sulfate half-ester	50 – 66	≈ 10.5	Evaporative concentration & UV crosslinking	–	–	–	–	Photonic materials	63
PSA, CNC	Sulfate half-ester	1.25 – 24.3	–	Free radical polymerization	–	–	–	7 – 32 x wet	Wound dressing	52,53
PAM, CNC	Sulfate half-ester	0 – 6	0 – 0.96	Free radical polymerization	–	≈ 5.5 – 33.8	6.1 x	–	–	60
PAM, CNC	Sulfate half-ester	0 – 50	0 – 0.25	Homogenization, casting	–	–	–	48.4 – 1457.7 x dry	Dye adsorption	61
PAM, PEG, CNC	Sulfate half-ester	–	0.1 – 3	Free radical polymerization	–	≈ 1 – 100	–	≈ 34 – 44 x dry	–	62
PNIPAM, clay, CNC	Sulfate half-ester	0 – 8	0 – 0.8	Freezing, free radical polymerization	6 – 15	–	2.5 x	30 – 45 x dry	Drug delivery	54
PNIPAM, CNC	HCl hydrolyze or AA (chem.)	0 – 5	0 – 0.25	Free radical polymerization, frozen	–	–	–	12.5 – 20 x dry	–	56
POEGMA, CNC	Sulfate half-ester	0 – 30.9	0 – 4.95	Coextrusion (injectable)	1.1 – 39	6 – 325	G' = 35.5 x	5 – 20 x dry	Biomedical	57
PAM, CNC	Sulfate half-ester	0 – 45	0 – 3.2	Freeze dried, free radical polymerization	≈ 2.5 – 7.5	–	≈ 3 x	≈ 20 – 30 x wet	–	83
PEG, CNC	Sulfate half-ester	–	0 – 1.13	UV polymerization	≈ 20 – 60	7.5 – 31.1	E = 4.1 x	4.4 – 13 x dry	Biomedical	65
Cyclodextrin-g-CNC, Pluronic	Polymer-grafted	0 – 5.96	–	Homogenization	≈ 1 – 30	–	–	–	Drug delivery	67
Pluronic, CNC	Sulfate half-ester	–	0 – 1.7	UV polymerization	–	≈ 0.3 – 100	–	–	–	66
CNC, agarose	Sulfate half-ester	20	1	Homogenization, reflux	–	–	–	–	Anisotropic materials	74
Agarose, CNC	Sulfate half-ester	0 – 50	0.02 – 0.32	Homogenization	0.08 – 0.83	–	10.4 x	–	–	75
CNC, chitosan-g-PAA	HCl hydrolyzed	0 – 10	–	Free radical polymerization, oven dried	–	–	–	381 – 486 x dry	Drug delivery	84
Gelatin, CNC	Sulfate half-ester	0 – 5	0 – 0.5	Homogenization with cross-linker, heat dried	–	–	–	6 – 12 x wet	–	72
CNC, gelatin	Sulfate half-ester	0 – 25	–	Homogenization, casting, drying	≈ 0.1 – 0.5	–	≈ 5 x	4.17 – 7.17 x wet	Drug delivery	73
Alginate, CNC	Sulfate half-ester	0 – 66	0.2 – 2	Homogenization, extruded into beads	–	–	–	–	Adsorbent	68,69
Alginate, CNC	Cationic (HPTMAC)	10	0.15	Ionic cross-linking	–	–	–	≈ 1.50 x wet	Drug delivery	70
Alginate, gelatin, CNC	Sulfate half-ester	0 – 2	0 – 0.04	Ionic cross-linking, injectable	–	50 – 92	1.84 x	–	Tissue engineering	71
Carrageenan CNC, FeCl ₂	Sulfate half-ester	5 – 30	0.1 – 0.6	Homogenization	–	–	–	–	Magnetic materials	85
CMC, HEC, CNC	COOH	0 – 40	0 – 0.8	Homogenization	–	10 – 30	–	≈ 1 – 4 x dry	–	80
Regenerated cellulose, CNC	Sulfate half-ester	0 – 50	0 – 2	Homogenization	≈ 1 – 3	–	≈ 3 x	≈ 6.5 – 10 x dry	Biomedical	79
HPC or CMC, CNC	Sulfate half-ester/COOH	0 – 18.6	0 – 1	Homogenization with cross-linker	–	≈ 39 – 49	1.3 x	≈ 120 – 170 x dry	–	76
CNC, MC	Sulfate half-ester	0 – 78	0 – 3.5	Homogenization	0.001 – 0.075	–	–	–	–	78
Cellulose, CNC	Sulfate half-ester	–	3 – 5	Homogenization, casting, vacuum dried	–	0.4 – 0.55 GPa	1.38 x	–	Energy storage	77
Cellulose, CNC	Cationic (HPTMAC)	0 – 50	0 – 3.5	Homogenization (injectable)	0.007 – 1.3	–	≈ 200 x	–	Drug delivery	86

^aSection entries are ordered and shaded by main polymer network component (PVA, yellow; polyacrylamides/polyacrylates, blue; PEG-base polymers, orange; natural polymers, grey; cellulose derivatives, green) and then chronologically. The “–” indicates information was not provided in the publication. “Swelling” is generally calculated from the mass of water uptake over time and compared to either the dry or initially hydrated (wet) hydrogel mass. ^bPolymer acronyms used: AA (acrylate), Ac (acrylate), CB[8] (curcubit [8] uril), CMC (carboxymethyl cellulose), HPTMAC (hydroxypropyl trimethylammonium chloride), HEC (hydroxyethyl cellulose), HPC (hydroxypropyl cellulose), HPG (hydroxypropyl guar), LBG (locust bean gum), MC (methyl cellulose), PAM (polyacrylamide), DAEMA (dimethylaminoethyl methacrylate), PEG (polyethylene glycol), PHEMA (polyhydroxyethyl methacrylate), PNIPAM (poly(*N*-isopropylacrylamide)), POEGMA (poly(oligoethylene glycol methacrylate)), PSA (poly(sodium acrylate)), PVA (poly(vinyl alcohol)), and PVA-MV (poly(vinyl alcohol-methyl viologen)).

charge densities largely designed to reduce energy input needed for fibrillation and to enhance colloidal stability. Without chemical treatment, small anionic surface charges are commonly reported and are attributed to residual hemicelluloses making CNFs strictly more “lignocellulosic” than CNCs which are pure cellulose. These long fibrils have a lower degree of crystallinity than CNCs (70% vs >85%)³⁰ and form physically entangled networks at extremely low concentrations (<1 wt %).

Given the high strength, dimensional anisotropy, and natural sourcing of both CNCs and CNFs, the use of nanocelluloses as a functional and renewable reinforcing agent for polymer composites has attracted significant research interest. However, the charged nature of the interface of nanocelluloses, coupled with their tendency to aggregate in more hydrophobic solvents or media, has made the efficient dispersion of CNCs/CNFs in most conventional polymeric materials challenging (although several surface modification techniques have been reported to improve such dispersions).⁶ Conversely, this charged interface of CNCs/CNFs is highly beneficial in their potential use as reinforcing agents or even building blocks of hydrogels, highly hydrated physically or chemically cross-linked three-dimensional networks that may encompass a wide range of chemical compositions and structural forms.³¹ The charge can be used to both stabilize the reinforcing agent in the aqueous precursor solution (promoting high dispersibility of the filler in the hydrophilic matrix) and interact with other gel components via electrostatic or hydrogen bonding or other means to lower the effective threshold for CNC/CNF gelation or tune the physical properties of the resulting hydrogels. In particular, tailoring the swelling and mechanical properties of hydrogels to the needs of a specific application is essential to create versatile and high performance functional materials, and the formation of high modulus hydrogels (aside from specific composition combinations, such as double network hydrogels)³² remains one of the barriers to the effective translation of hydrogels to practical applications. Furthermore, the apparent biological inertness of CNCs/CNFs across a range of studies^{11,33–36} makes them attractive in applications such as tissue engineering scaffolds, wound healing materials, and drug delivery systems, all of which are particularly well-served by the high water content, porosity, and typical cytocompatibility of hydrogels.^{37,38}

Similar benefits are offered by nanocelluloses in the preparation of aerogels, highly porous lightweight materials with an average pore size in the tens of nanometers produced by sublimating the liquid component from conventional gels (often via freeze-drying or critical point drying).³⁹ The potential of aerogels to exhibit well-defined pore morphologies and selective sorption properties has attracted considerable interest for sorbents, insulating materials, biomimetic substrates, and templates for other functional materials. Relative to other aerogel building blocks, nanocelluloses offer the potential for excellent interfacial chemistry control (essential for the selectivity of any sorbent), enhanced mechanical properties (currently, a key translational barrier of other aerogels in, for example, higher pressure applications), and extremely high specific surface areas (due to their anisotropic structure), in addition to their beneficial renewable origins.

The wide range of processing methods for preparing nanocellulose hydrogels and aerogels also adds to the broad interest and application of these materials. Hydrogels are commonly prepared via addition of CNC/CNF to a polymer precursor solution, followed by free radical polymerization to

chemically cross-link the main polymer component around the nanocellulose. In some cases (e.g., supramolecular gels, gels with chemically orthogonal precursors, CNF-based gels), simple mixing/homogenization of precursor materials causes gelation, leading to hydrogels with interesting properties including injectability and self-healing. Freeze–thaw cycling, a process where precursor material is rejected from growing ice-crystals (freeze) creating water-filled pores (thaw), can also be used to create strong physical hydrogel networks. On the other hand, aerogels are traditionally prepared via freeze-drying (low temperature, low pressure) or critical point drying (CPD, high temperature, high pressure) of a precursor material or gel. In both cases, the aqueous gel phase is replaced with gas (air), by transitioning through either a solid (freeze-drying) or supercritical (CPD) regime. Freeze casting is another common preparation method acting on a similar principle to freeze-drying; when a temperature gradient exists during the freezing process, ice crystallites tend to grow directionally, creating anisotropic aerogels upon drying. These aerogels may then be used as is or rehydrated to form hydrogels with new pore morphologies and physical properties.

This Review highlights the developments and applications of nanocellulose hydrogels and aerogels with extensive tables summarizing the current literature. The preparation methods, physical, chemical, and mechanical properties, and intended applications are provided wherever possible. Hydrogels are reviewed first, divided between CNC and CNF-derived hydrogels with subsections highlighting whether the incorporation of nanocellulose is physical (i.e., entanglement, aggregation, entrapment, supramolecular interactions, or physical adsorption) or through the introduction of new covalent bonds between nanocellulose and the polymer matrix (i.e., chemical cross-linking). The second section reviews aerogels based on CNCs and CNFs, divided in the tables by “nanocellulose-only” aerogels and hybrid aerogel materials and including those that contain other functional polymers and nanoparticles for advanced applications such as energy storage and water purification.

2. HYDROGELS

2.1. “CNC-Only” Hydrogels. Due to their rigidity and mechanical strength, CNCs act as effective fillers in hydrogel nanocomposites. However, their limited ability to “entangle” generally causes CNCs alone to be ill-suited as single-component gels. This is reflected in the fact that there are very few published examples of “CNC-only” hydrogels; these are presented in Table 1 along with their method of preparation and storage moduli. As mentioned, CNCs above 10 wt % will transition into an aggregated gel-like phase,⁴⁰ or gels can be prepared by destabilizing CNC suspensions with salt or acid or cross-linking CNCs with multivalent ions or chemical bonds.^{41,42} In practice, the CNC concentration at which a gel is formed may be manipulated via changing the solution conditions, surface modifications, and/or adding adsorbing or nonadsorbing water-soluble polymers.^{27,41–44} Chau et al. have shown that increasing the ionic strength of CNC suspensions via salt addition suppresses electrostatic repulsion and leads to dominant attractive forces (e.g., van der Waals and hydrogen bonding) and consequent gelation.⁴² Specifically, increasing the charge number and the radii of the ions both led to increases in gel stiffness.⁴² Regardless of the cation type added (Na^+ , Mg^{2+} , Al^{3+} , Ca^{2+} , Sr^{2+}), the sol–gel transition was determined to occur around 1.5 wt %, roughly 1 order of magnitude lower than the

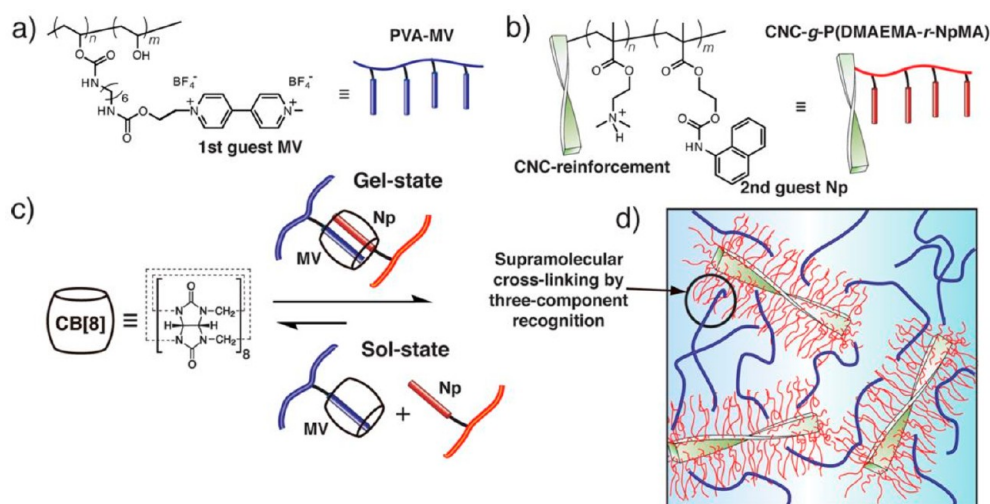


Figure 1. Schematic of a dynamic three component supramolecular hydrogel composed of (a) PVA with methyl viologen (PVA-MV), (b) CNCs containing copolymer grafts of protonated dimethylaminoethyl methacrylate and naphthyl methacrylate (CNC-g-P(DMAEMA-r-NpMA)), and (c) the host cucurbit[8]uril (CB[8]). (d) Copolymer brushes were grafted to CNC surfaces through surface-initiated atom transfer radical polymerization, which were then combined with PVA-MV and CB[8] and centrifuged to form a homogeneous rubbery material with CNCs well-dispersed throughout the network. Although the modification of CNCs and PVA may be limiting on a large scale, this physical gel reports the highest modulus for a PVA-CNC gel to date. Reprinted with permission from ref 50, copyright 2016 John Wiley & Sons, Inc.

onset of gelation for pure CNC suspensions investigated by Ureña-Benavides et al.⁴⁰

Physical stimuli can alternatively be used to drive gelation. Way et al. functionalized CNCs with either carboxyl or amine groups to achieve pH-responsive gelation in “CNC-only” hydrogels.⁴¹ In the case of amine functionalized CNCs, at high pH, the amine groups have a neutral charge, allowing the attractive forces to dominate; in contrast, at low pH, the amine groups are protonated, facilitating electrostatic repulsion between CNCs and inhibiting aggregation. The opposite trend is observed with the carboxylic acid-modified CNCs, whereby the storage modulus (G') of a 2.7 wt % CNC suspension increases by 3 orders of magnitude from pH 11 to pH 1.⁴¹ Way et al. went on to show that these pH responsive CNCs may be incorporated into a poly(vinyl acetate) matrix to achieve mechanically adaptive thin films.⁴¹ Thermal initiation of CNC gelation was also demonstrated, although the mechanism appears to be linked primarily to temperature-induced chemical modification rather than physical self-assembly. Lewis et al. demonstrated that autoclaving dilute suspensions of sulfated CNCs above 80 °C resulted in gel formation, attributed to CNC desulfation that drastically reduced the CNC colloidal stability.⁴⁵ Dorris and Gray similarly demonstrated that CNC desulfation was necessary to induce gel formation in dilute CNC suspensions in a glycerol/water mixture.⁴⁶

Polymer adsorption to the surface of CNCs can also be used to drive gel formation, although the resulting gels are not strictly CNC-only materials. Hu et al. showed that 3 wt % CNC suspensions can be “pushed” to form a nematic gel via polymer adsorption with 0.2 wt % nonionic polysaccharides (hydroxyethyl cellulose, locust bean gum, or hydroxypropyl guar). Adsorption effectively increases the CNC volume fraction and shifts the liquid crystalline phase diagram to lower CNC concentrations.²⁷

2.2. Hydrogels with Physically Entrapped CNCs. To date, the majority of the CNC hydrogel research has focused on the straightforward physical incorporation of CNCs as filler/reinforcing agents within polymeric hydrogel networks. Table 1

highlights this work, comparing the preparation methods, physical properties, and intended applications of CNC hydrogels with common network polymers including poly(vinyl alcohol) (PVA), polyacrylamides (Pam), poly(meth)acrylates, poly(ethylene glycol) (PEG)-based materials, water-soluble polysaccharides, and other natural polymers such as alginate and gelatin. Common preparation methods include simple homogenization and physical entanglement of polymer networks, free radical polymerization of polymeric species within CNC suspensions, UV/ion mediated cross-linking of the polymeric network around the CNCs, and cyclic freeze-thaw processing. CNCs are generally physically incorporated into hydrogels at low concentrations, with the majority of studies employing concentrations of 0.1–5 wt % based on total swollen gel weight; such incorporations have resulted in up to 35-fold reinforcement of the shear storage modulus of the hydrogels (G' as high as 60 kPa) and absorption of up to 1500 times their dry mass in water (Table 1).

PVA networks have been utilized as biomaterials in many cases due to their relative biocompatibility, biodegradability, stability, and high elasticity. Gonzalez et al., Abitbol et al., and Han et al. showed moderate increases in PVA-based hydrogel mechanical properties upon CNC loading compared to the relatively weak PVA-only controls.^{47–49} McKee et al. has shown among the most significant enhancements in the stiffness of self-healing hydrogels by utilizing three-component recognition-driven supramolecular cross-links, as shown in Figure 1.⁵⁰ Increasing the CNC concentration from 0.1 to 0.2 wt % resulted in an increase in the storage modulus of the resulting gels from 3.8 to 14.3 kPa, the latter representing the highest modulus of all PVA-based networks investigated.⁵⁰

Polyacrylamide and acrylate-based networks with CNC fillers have been studied extensively due to their relative ease of (co)polymerization and cross-linking.^{51–61} Thermoresponsive poly *N*-isopropylacrylamide (PNIPAM) polymer networks have also been investigated with CNC fillers, showing characteristic temperature dependent swelling properties along with tunable mechanics.^{54,56} In most cases, small

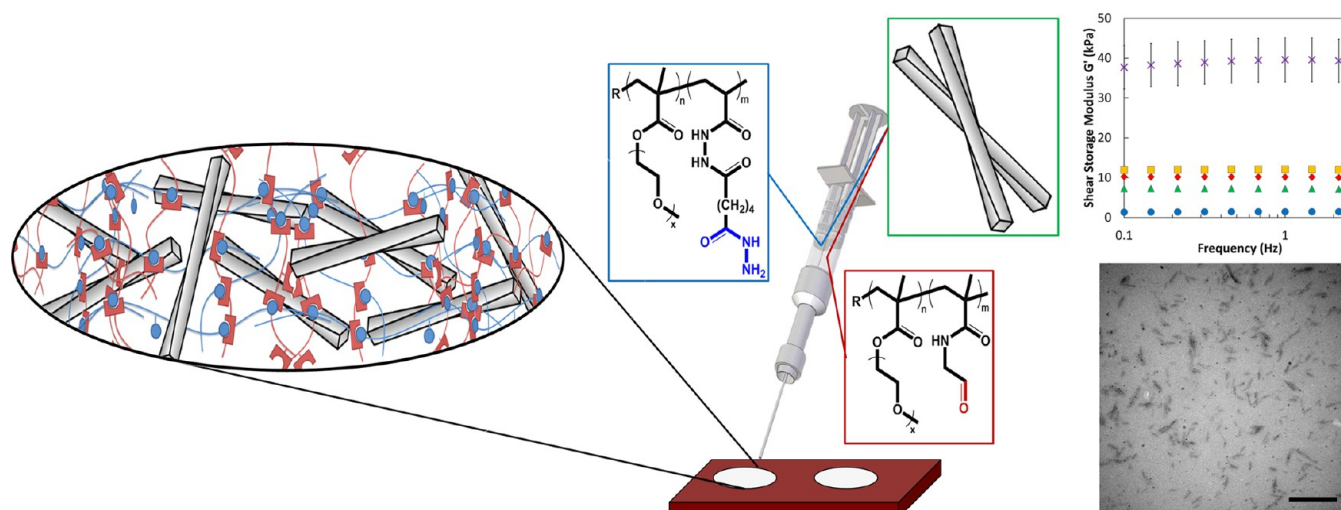


Figure 2. A schematic illustration of injectable CNC reinforced poly(oligoethylene glycol methacrylate) (POEGMA) nanocomposite hydrogels. (Top right) Increasing the CNC concentration from 0 (○) to 0.2 (Δ), 0.96 (◇), 1.65 (□), and 4.95 wt % (×) led to successive increases in the shear storage modulus from 1.1 to 39 kPa, indicating that this hydrogel platform may be relevant for a wide range of tissue engineering applications spanning a range of mechanical strengths. (Bottom right) CNCs are well-dispersed in the POEGMA network, as shown through SEM imaging (scale bar = 1 μm). Adapted from ref 57. Copyright 2016 American Chemical Society.

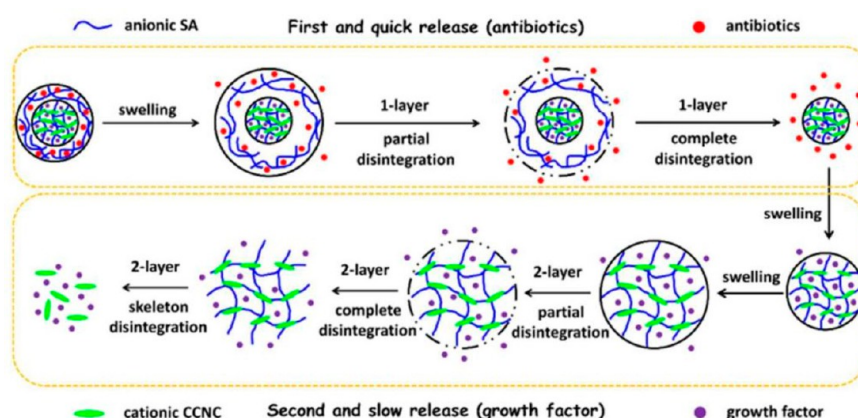


Figure 3. Proposed complexing drug release mechanism for double-membrane anionic sodium alginate (SA)/cationic CNC (CCNC) hydrogel beads at pH 7.4. The hydrogel beads first undergo partial swelling and degradation of the alginate shell, facilitating rapid drug release of a model antibiotic, followed by a slower swelling and degradation of the ionically cross-linked anionic alginate/cationic CNC core, sustaining drug release over time of a model growth factor. Reprinted from ref 70. Copyright 2016 American Chemical Society.

molecule cross-linkers such as methylene bis(acrylamide) (MBA) have been used to create a chemically cross-linked polymer network with physically incorporated CNCs.^{51–53,56,58–60} Alternatively, Zhou et al. used heat treatment to obtain physical polyacrylamide hydrogels, which when filled with CNCs proved to be highly effective for the absorption of methylene blue dye.⁶¹ Yang et al. further demonstrated how CNC surface charge and aspect ratio affected reinforcing ability in polyacrylamide hydrogels, showing that a higher surface charge density led to better dispersibility while a higher aspect ratio contributed to increased mechanical properties.⁶² The MacLachlan group has demonstrated UV curable polyacrylamide nanocomposite hydrogels with chiral nematic ordering upon solvent evaporation and concentration of the CNC suspension.⁶³ While CNCs have a tendency to form chiral nematic liquid crystals above a critical concentration, this is the only example of a chiral composite hydrogel with CNCs and it is somewhat surprising that the acrylamide monomer and the polymerization

process around the CNCs did not disrupt the CNC self-organization. These hydrogels display solvent/pH responsive swelling properties, which changes the overall hydrogel iridescence and could prove useful for photonic and optoelectronic material design.⁶³

Work has been done in our group utilizing hydrazide-aldehyde “click” chemistry to create injectable methacrylate-based and polysaccharide-based hydrogels without the need for additional small molecule cross-linkers.^{57,64} In the case of polysaccharide-based hydrogels, the CNCs were well-distributed and did not strongly interact with the polymers, leading to significant but relatively modest enhancements in gel elasticity. However, with poly(oligoethylene glycol methacrylate)-based hydrogels, we observed a strong physical interaction between both aldehyde and hydrazide-modified polymers onto the surface of CNCs, leading to hydrogel networks with well-dispersed CNCs and drastically improved mechanical properties versus the polymer-only controls. The incorporation of 4.95 wt % CNCs led to a greater than 35-fold increase in the gel

Table 2. Properties of Polymer Hydrogel Networks with Chemically Entrapped CNCs^a

Materials ^b	CNC Surface Chemistry	wt% CNC (vs. Polymer)	wt% CNC (Total in Gel)	Hydrogel Preparation Method	Storage Modulus (G', kPa)	Compressive Modulus (E, kPa)	Reinforcement	Swelling (pH 7 in H ₂ O)	Intended Application	Ref
Chemical Entrapment of CNCs into Hydrated Polymer Networks										
PVA, CNC	COOH	–	3.88	Homogenization, freeze-thaw	10–20	–	–	–	Biomedical	105
PVA, CNC-g-AA-c-AM-c-AMPS	Polymer grafted	50–90	0.5	Free radical polymerization	–	–	–	≈ 10 x dry	Drug delivery	106
Starch, PVA, CNC	HCl hydrolyzed	0–10	0–5	Homogenization, emulsion polymerization	–	–	–	–	Drug delivery	104
PMVE-co-MA, PEG, CNC	Sulfate half-ester	0–100	–	Solution casting, heat cured	–	0.09–3.9 x 10 ⁶	–	–	Biomedical	107
PAM, CNC	Sulfate half-ester	0–9.3	0–0.93	Free radical polymerization	≈ 0.5–2.4	–	4.8 x	≈ 20–45 x dry	–	87,88
PAA, CNC	Silylated	0.5–5	0.1–1	Free radical polymerization	10–300	–	–	≈ 300–550 x dry	–	92,93
PAM, CNC	Silylated	0–3	0–0.9	Free radical polymerization	≈ 1–100	3.1–21.5	E = 6.9 x	≈ 25–35 x dry	Biomedical	90,91
PNIPAM, CNC	Sulfate half-ester	0–50	0–5	Free radical polymerization	–	–	–	≈ 5–20 x dry	–	89
PDMAM, CNC	Silylated	0–0.5	0–0.08	UV polymerization	8–100	4.7–22.7	E = 4.8 x	–	–	94
PAM, CNC-g-GMA	Polymer grafted	0–50	0–5	Free radical polymerization	–	110–330	10.3 x	9–24 x dry	Biomedical	95
PAM, CNC	Polymer grafted (acylated)	–	–	Free radical polymerization	–	3.1–47.4	0.28 x	10.1–36.7 x dry	–	96
POEGMA, CNC	Aldehyde	20–50	0.5–3.5	Freeze casting	–	0.8–80.2	–	–	Biomaterials	97
CMC, CNC dextran	Aldehyde	0–25	0–1	Coextrusion (injectable)	3–6.7	–	1.4 x	2.5–3 x wet	Tissue engineering	64
Gelatin, CNC	Aldehyde	–	–	Homogenization, RT drying	12–32	–	1.5 x	8–13 x dry	Biomedical	101
Xylan, CNC	Sulfate half-ester	20	1.25	Freeze casting	–	3.7–285	–	–	–	100
HA, CNC	Aldehyde	0–33	0–0.5	Coextrusion (injectable)	107–152	–	1.35 x	≈ 1.5–2 x wet	Tissue engineering	99
Collagen, Gelatin microsphere CNC	Sulfate half-ester	0–10	–	Homogenization, freeze drying	–	–	–	16–20 x dry	Angiogenic tissue engineering	103
Gelatin, CNC	Maleimide	–	0–6.3	Homogenization	≈ 3	–	–	22–32 x dry	–	102

^aSection entries are ordered by main polymer component and then chronologically. The “–” indicates information was not provided in the publication. ^bPolymer acronyms used: AA (acrylate), AM (acrylamide), AMPS (2-acrylamido-2-methyl-1-propanesulfonic acid), CMC (carboxymethyl cellulose), GMA (glycidyl methacrylate), HA (hyaluronic acid), MA (maleic acid), PAM (polyacrylamide), PDMAM (poly(*N,N*-dimethacrylamide)), PEG (poly(ethylene glycol)), PMVE (poly methyl vinyl ether), PNIPAm (poly(*N*-isopropylacrylamide)), POEGMA (poly(oligoethylene glycol methacrylate)), and PVA (poly(vinyl alcohol)).

shear storage modulus (Figure 2), which was accompanied by a decrease in the gelation time and equilibrium swelling ratio and an increase in cell adhesion, useful in biomedical applications.⁵⁷ Other synthetic polymers including PEG⁶⁵ and Pluronic,^{66,67} have also been shown amenable to CNC incorporation. For example, consistent increases in gel mechanical strength were observed by Yang et al., who utilized UV initiated cross-linking to form the polymer network in the presence of CNCs.^{65,66} Lin and Dufresne also designed supramolecular hydrogels (similar to McKee et al.⁵⁰) based on β -cyclodextrin grafted CNCs and Pluronic that showed an increase in structural and thermal stability as well as a more gradual drug release profile than controls without CNCs.⁶⁷

As many CNC-containing hydrogels target biological or biomedical applications, there has been an increasing amount of research on natural polymer-based hydrogels, primarily based on alginate,^{68–71} gelatin,^{71–73} and agarose.^{74,75} The Tam group focused on the use of CNC-containing alginate beads as hydrogel absorbents for natural dyes, showing that increasing the CNC concentration causes an increase in the removal efficiency of methylene blue.^{68,69} Other groups have used similar alginate-based hydrogels for the delivery of drugs, cells, or other bioactive molecules; for example, Lin et al. created double-membrane hydrogels with an alginate shell to facilitate rapid drug release and an anionic alginate/chemically modified

cationic CNC core for prolonged release.⁷⁰ Such a structure enabled dual release of ceftazidime hydrate, which was loaded in the shell and rapidly released in vitro within the first 3 days, and epidermal growth factor, which was loaded in the core and slowly released from day 4 to day 12 (Figure 3).⁷⁰

Both the hydrogel and CNC phases can be manipulated to facilitate desirable changes in gel properties. From the hydrogel perspective, CNC-reinforced interpenetrating networks have been introduced to exploit the inherent strengthening effect as well as potential partitioning benefits of interpenetrating gels in a CNC composite. For example, Wang et al. demonstrated an interpenetrating network of alginate and gelatin to create an injectable hydrogel to facilitate cell migration and transportation of biomolecules.⁷¹ The physical incorporation of CNCs increased the gel compressive modulus and stability against degradation, while having little effect on cell viability versus the CNC-free controls.⁷¹ Le Goff et al. showed that CNCs have a significant reinforcement effect of more than 1 order of magnitude on the storage modulus of agarose hydrogels.⁷⁵ Alternately, alignment of CNCs within the hydrogel can promote anisotropic hydrogel properties. For example, Osorio-Madrado et al. subjected CNC/agarose nanocomposite gels to uniaxial stretching under varying humidities, and CNC orientation was monitored via X-ray diffraction. Lower relative humidity and cyclic hydrogel

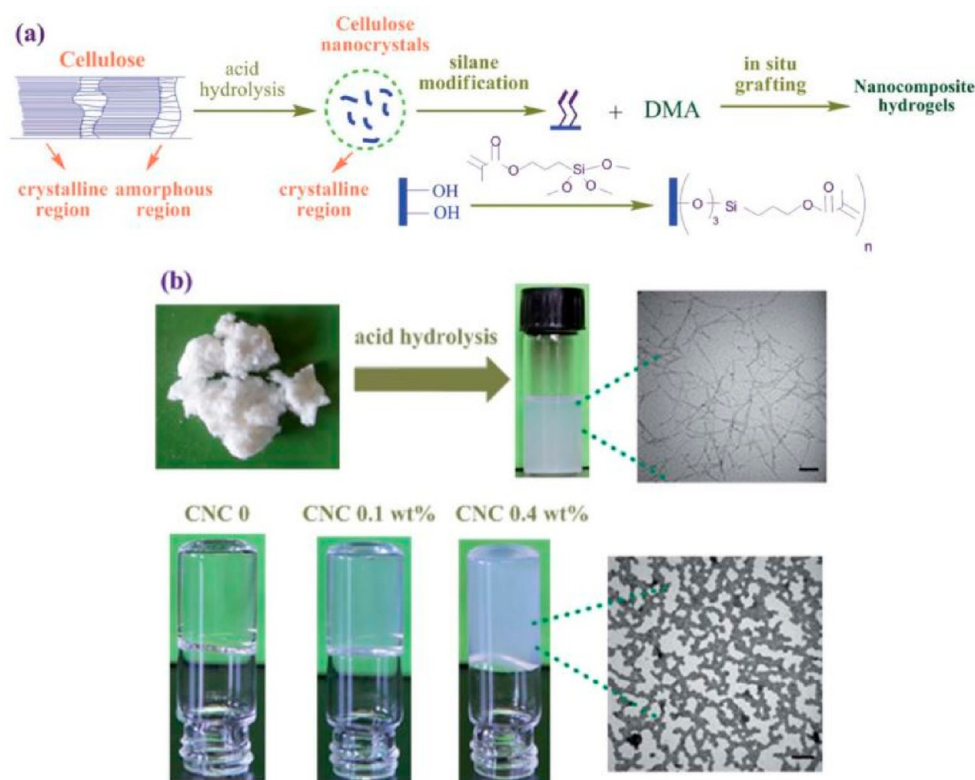


Figure 4. (a) Schematic representation of silane modification of CNCs to create nanocomposite hydrogels. (b, top panel) TEM images show CNCs after acid hydrolysis and (b, bottom panel) after free radical polymerization of poly(*N,N*-dimethylacrylamide) (DMA) to form percolating gel networks. Reprinted with permission from ref 94, copyright 2014 Royal Society of Chemistry.

rehydration led to the highest level of CNC reorientation in the straining direction, proving an effective method for creating anisotropic hydrogel nanocomposites.⁷⁴

Many researchers have focused on creating “all-cellulose” gels with widely available and biocompatible cellulose derivatives.^{76–80} Notably, McKee et al. utilized thermoresponsive methylcellulose networks, which formed hydrogels at elevated temperature and were successfully reinforced by CNCs with the storage modulus increasing by nearly an order of magnitude upon addition of 3.5 wt % CNC.⁷⁸ The ability of CNCs to effectively reinforce all natural or all-cellulose polymer networks drastically broadens the potential for CNCs in advanced sustainable/green chemistry applications.

2.3. Hydrogels with Chemically Entrapped CNCs.

Table 2 summarizes the various approaches that have been reported for the chemical incorporation of CNCs into hydrated polymer networks. In all cases, new covalent bonds are formed between the CNCs and the surrounding hydrogel phase, mandating strategic modification of the surface chemistry and functionality of the CNCs. In some examples, CNCs are the polymer cross-linkers themselves; in others, CNCs are cross-linked to a networked polymer matrix. Loadings of CNCs in chemically cross-linked hydrogels are similar to physical gels at 0.1–6 wt % (based on the total hydrated gel weight) but can typically achieve significantly higher G' values (up to 300 kPa modulus for the strongest systems reported). If similar polymer systems and loadings of physically vs chemically incorporated CNCs are directly compared, the covalently cross-linked hydrogels are normally stronger and swell less (Table 2). Cross-linked CNC-based hydrogels are therefore highly versatile and mechanically stable, presenting a platform for a

wide variety of biomedical applications as demonstrated with these highlighted examples.

The most studied polymer matrix investigated with chemically incorporated CNCs has been polyacrylamide/polyacrylate-based networks. For the most part, chemical incorporation of CNCs has been accomplished through surface modification of CNCs to create covalent cross-linking sites (commonly, polymer grafts, silyl groups, or carboxyl or aldehyde groups), although some researchers have reported that free radical polymerization can occur from unmodified CNC surface hydroxyl groups using small molecule cross-linkers (MBA).^{87–89} In one series of work, Yang and colleagues demonstrated the silane modification of CNCs to covalently incorporate CNCs into polyacrylamide hydrogels via free radical polymerization.^{90–94} This free radical polymerization proceeds by nucleating silane-modified CNCs in acrylamide monomer solution which leads to aggregation into larger clusters through physical interactions to form a percolated network, as depicted in Figure 4.⁹⁴ These networks are mechanically tough and stretchable, possessing a more efficient energy dissipation mechanism than the control acrylamide hydrogels, and broaden the potential applications of CNC nanocomposites.⁹⁴

When covalent cross-linking between polymers and modified CNCs is utilized, it provides the advantage of hydrogels having drastically increased (ca. 1 order of magnitude) mechanical properties versus control hydrogels cross-linked with small molecules, even with less than 1 wt % CNCs.⁹⁰ As mentioned above, in most cases, the mechanical improvement of hydrogels with chemically incorporated CNCs surpasses their physically incorporated counterparts. For example, Atifi et al. grafted

glycidyl methacrylate onto the surface of CNCs and then initiated free radical polymerization from CNCs;⁹⁵ a significant reinforcing effect (almost 2-fold increase in compressive modulus at 50 wt % CNC based on polymer concentration) was noted when surface-modified CNCs were used to prepare gels instead of unmodified CNCs and small molecule cross-linkers. Yang et al. looked at the effect of grafted carbon chain length on acylated CNCs on the properties of polyacrylamide composite hydrogels.⁹⁶ In general, longer carbon chain lengths led to significant increases in hydrogel tensile strength and water swelling ratio, creating softer and more elastic gels.⁹⁶

Our group has recently used aldehyde-modified CNCs with hydrazide-modified poly(oligoethylene glycol methacrylate) (POEGMA) to create tailorable hydrogels with reversible hydrazone cross-links.⁹⁷ Furthermore, we created anisotropic hydrogels by directional freeze-casting which led to materials with tunable pore morphologies depending on the total mass of CNCs (0.5–3.5 wt %) and ratios of CNC to POEGMA (1:1–1:5), as shown in Figure 5. Hydrogel mechanical properties and

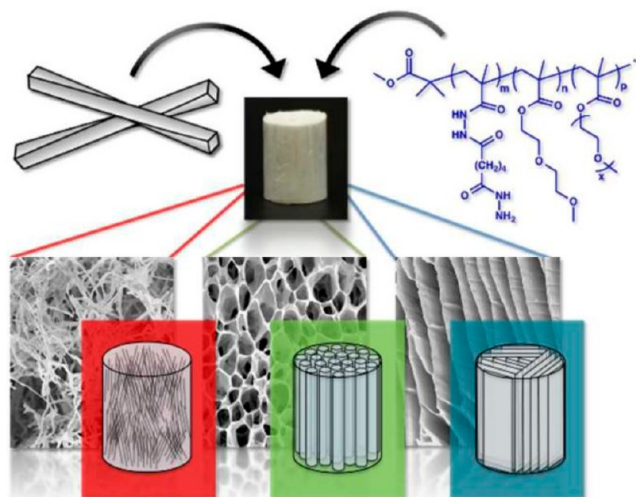


Figure 5. CNC–POEGMA nanocomposite hydrogels formed through hydrazone cross-linking and subsequent directional freeze casting. SEM reveals hydrogels with highly tunable pore morphologies (fibrillar, columnar, lamellar) and anisotropy. Reprinted from ref ⁹⁷. Copyright 2016 American Chemical Society.

swelling behavior exhibited directionally dependent properties, demonstrating a higher compressive modulus and lower swelling ratio in the direction of ice crystal growth; this suggests that these materials may function as effective biomimetic scaffolds for oriented tissues.^{97,98}

Reports combining natural polymer networks and CNCs are also prevalent, with slightly more focus on chemical incorporation compared to physical incorporation to create mechanically robust, sustainable hydrogels. Previous work in our group demonstrated injectable hydrogels of hydrazide-functionalized carboxymethyl cellulose (CMC) and aldehyde-functionalized dextran reinforced with aldehyde-modified CNCs.⁶⁴ We showed that these hydrogels contained evenly dispersed CNCs and displayed increased stability and mechanical strength versus unmodified CNC control gels; however, CNC loadings above 0.4 wt % seemed to interfere with the CMC/dextran cross-linking. Thus, a maximum in mechanical properties was observed.⁶⁴ In that work, no significant cytotoxicity was observed on mouse fibroblast

cells, indicating that these materials may be of potential interest for biomedical applications.⁶⁴ Domingues et al. also utilized aldehyde–hydrazide chemistry to create injectable and chemically cross-linked hyaluronic acid hydrogels reinforced with aldehyde-modified CNCs.⁹⁹ The level of reinforcement, decreased swelling, and increased resistance to degradation were comparable to results found in our group. Moreover, human adipose derived stem cells demonstrated an ability to spread within the hydrogel network and exhibited pronounced proliferative activity.⁹⁹

Various cross-linking chemistries have been explored to link CNCs with natural polymers to form reinforced hydrogel structures. Köhnke et al. cross-linked aldehyde-modified xylan with unmodified CNCs through hemiacetal bonds, which are less stable than some of the cross-linking chemistries described previously.¹⁰⁰ Through freeze casting, they were able to obtain anisotropic hydrogels with lamellar pore morphologies. Similarly, Dash et al. found that increasing the aldehyde content in aldehyde-modified CNCs led to an increased imine cross-link density in CNC/gelatin hydrogels; this was accompanied by reduced water uptake and slightly increased mechanical properties versus neat gelatin.¹⁰¹ Finally, García-Astrain et al. demonstrated Diels–Alder “click” cross-linking between furan-modified gelatin and maleimide-modified CNCs, showing reduced water uptake and stiffer networks when covalently linked CNCs were used,¹⁰² demonstrating the versatility of CNC modifications for creating chemically cross-linked hydrogels.

CNCs have also been incorporated into composite hydrogel structures to both improve mechanical performance and tune drug release. As proof of principle, Li et al. loaded basic fibroblast growth factor into biodegradable gelatin microspheres that were subsequently incorporated into porous glutaraldehyde-cross-linked collagen/CNC scaffolds in order to promote angiogenesis in skin tissue engineering applications. Interestingly, hydrogels with 5 wt % CNC (relative to collagen) exhibited a higher swelling ratio than pure collagen gels, attributed to an increased hydrophilicity of the gels.¹⁰³ The gelatin microspheres contributed to a sustained drug release profile and increased angiogenesis upon subcutaneous implantation in Sprague–Dawley rats.¹⁰³ Alternately, Mauricio et al. prepared an emulsion of vinyl-functionalized starch and CNCs which underwent free radical polymerization to form microgels on the order of 1 μm .¹⁰⁴ The addition of 10 wt % CNCs resulted in a more sustained release profile for vitamin B12, approximately 2.9 times slower than starch-only microgels.¹⁰⁴

2.4. “CNF-Only” Hydrogels. Due to their increased flexibility and propensity for entanglement, cellulose nanofibrils on their own lend themselves more readily to hydrogel formation than do CNCs. As such, there are many more publications on “CNF-only” hydrogels, as summarized in Table 3. Typical “CNF-only” hydrogels contain 0.05 to 6 wt % CNFs, with G' values ranging up to 100 kPa having been reported. Pääkkö et al. were the first to show that CNF gels were possible at concentrations as low as 0.125 wt % following an enzymatic treatment and homogenization of bleached sulfite pulp,²⁸ 2 orders of magnitude concentration lower than that required for CNC gelation⁴⁰ (albeit with very weak mechanical properties, prompting research focused on destabilizing the electrostatic repulsion/DLVO interactions of individual CNF fibers to better control gel properties).^{108–110} Fall et al. showed that, by adding salt or lowering the pH, CNF suspensions readily formed gels

Table 3. “CNF-Only” Hydrogels and Polymer Hydrogels with Physically and Chemically Entrapped CNFs^a

Materials ^b	CNF Surface Chemistry	wt% CNF (vs Other)	wt% CNF (Total in Gel)	Preparation Method	Storage Modulus (G', kPa)	Compressive Modulus (E, kPa)	Reinforcement	Swelling (pH 7 in H ₂ O)	Intended Application	Ref
“CNF Only” Hydrogels										
CNF	Unspecified, from Domsjö	100	0.125 – 5.9	Homogenization	0.001 – 100	–	–	–	–	28
CNF	COOH/Cationic (HPTMAC)	100	0.1	Homogenization	–	–	–	1.4 x wet	–	137
CNF	COOH	100	0.1 – 0.8	Homogenization/ pH reduction	≈ 0.1 – 20	–	–	–	–	109,138
CNF	Unspecified	100	10 or 0.8	Vacuum filter & NaOH treatment	–	–	–	12.1 – 13.4 x dry	–	114,115
CNF	Unspecified, from UPM Corp.	100	1.7	Homogenization (injectable)	0.01 – 0.02	–	–	–	Cell culture	116
CNF	COOH/aldehyde	100	0.05	Homogenization, uranyl cross-linking	–	–	–	–	Absorbent	121
CNF	Unspecified, from UPM Corp.	100	0.2	Homogenization	–	–	–	–	Drug release	139
CNF	COOH	100	1.27	Homogenization/ salt-bridging	3.4 – 32	–	–	–	Biomedical	117,119
CNF	Carboxy-methylated	100	0.14	Homogenization/ pH reduction	≈ 0.1 – 1	–	–	–	Templating	108,111
CNF	COOH/aldehyde	100	0.1	Homogenization / salt bridging	–	–	–	–	–	110
CNF	Carboxy-methylated	100	0.3	Hydrodynamic alignment, salt bridging	–	–	–	–	–	112
CNF	Unspecified, from UPM Corp.	100	1	Homogenization (injectable)	–	–	–	–	Radio-labelling	123
CNF	COOH	100	–	Homogenization/ salt bridging	–	–	–	–	Tissue culture	120
CNF	COOH or aldehyde	100	0.2 – 0.4	Homogenization/ 3D printing, salt-bridging	–	–	–	–	–	140
CNF	COOH	100	0.2 – 1.2	Homogenization/ salt-bridging	≈ 1 – 4	–	–	60 – 80 x wet	–	118
CNF	COOH	100	1	Sacrificial templating, centrifugation	–	–	–	–	Tissue engineering	122
CNF	Unspecified or COOH	100	1 – 10	Homogenization, wet spinning	–	–	–	–	–	113
Physical Entrapment of CNFs into Hydrated Polymer Networks										
PVA, Borax, CNF	Sulfate half-ester	30	1	Homogenization	2.35	22.5	–	–	Biomedical	49
Chitosan, PVA, CNF	COOH	–	1	Homogenization, extruded into beads	–	–	–	–	Lead absorbent	133
CNF, PVA, Borax	Unspecified, from UPM Corp.	20 – 60	–	Homogenization	–	28 – 97	–	2.3 – 11 x dry	Wound dressing	141
Alginate, PVA, CNF	Carboxy-methylated	–	0 – 1	Homogenization	0.01 – 0.2	4 – 15	G' ≈ 20x	–	Absorbent	135
NVP, CNF	Carboxy-methylated	–	0 – 1.6	Photo-polymerization	0.75 – 2.75	7.4 – 62.2	E = 3 – 8 x	1.1 – 5.7 x wet	Nucleus pulposus implant	124,125
CMC-g-PAM-co-PAA, CNF	COOH	0 – 20	0 – 2.5	Free radical polymerization	–	–	–	≈ 150 – 450 x dry	Absorbent	134
PEG, CNF	COOH	10 – 50	0.2 – 2	Ionic cross-linking	≈ 0.1 – 1	–	–	–	–	130
PNIPAM, CNF	COOH	0 – 10	–	Free radical polymerization	–	75 – 325	4.3 x	≈ 30 – 42 x dry	–	131
PEGMEMA-co-DMAm, CNF	COOH	20 – 100	0.26 – 1.3	Free radical polymerization	–	–	–	–	–	129
PAM, CNF	COOH	≈ 5	≈ 0.34	Photo-polymerization & cation addition	≈ 7 – 11	–	–	–	–	142
Collagen, CNF	Unspecified, from Domsjö	50 – 75	–	Solution casting	–	≈ 5000 – 14500	≈ 1.5 – 4 x	1.3 – 3.7 x wet	–	126,127
CNF, gelatin, chitosan	Unspecified, from Domsjö	70 – 90	–	Homogenization	–	1000 – 3000	–	10 – 17 x dry	Articular cartilage	128
Hemicellulose, CNF	COOH, aldehyde	70 – 100	0.1	Homogenization	–	9.8 – 107.8	–	≈ 5 – 600 x wet	Wound healing	143
Alginate sulfate, CNF	Unspecified, from CELLINK	50 – 80	1.36	Homogenization	14.6	–	–	–	Bio-inks	144
Chemical Entrapment of CNFs into Hydrated Polymer Networks										
CNF, PEI or PNIPAm	COOH	44 or 91	0.5	Freeze-thaw	–	–	–	–	–	132,145
CNF, (diamine cross-linker)	COOH/ aldehyde	–	–	Freeze dried, heat cured/salt addition	–	–	–	≈ 20 – 160 x dry	–	136
PMVE-co-MA, PEG, CNF	Unspecified, from Arbocel	20 – 60	1 – 3	Homogenization, heat cured	17 – 37	–	–	1.1 – 2.2 x wet	Biomedical	146
CNF, diamine cross-linker	COOH/ aldehyde	–	0.5 – 1	Homogenization, heat cured	≈ 0.03 – 1.1	≈ 1 – 3.2	–	≈ 35 – 85 x dry	–	147

^aSection entries are ordered by polymer component and then chronologically. The “–” indicates information was not provided in the publication.

^bPolymer acronyms used: CMC (carboxymethyl cellulose), DMAm (N,N-dimethacrylamide), HPTMAC (hydroxypropyl trimethylammonium chloride), MA (maleic acid), NVP (N-vinyl-2-pyrrolidone), PAA (poly acrylic acid), PAM (poly acrylamide), PEG (poly ethylene glycol), PEGMEMA (poly ethylene glycol methyl ether methacrylate), PEI (poly ethylenimine), PMVE (poly methyl vinyl ether), PNIPAm (poly N isopropylacrylamide), and PVA (poly vinyl alcohol).

via reduction in surface charge through interaction of counterion-driven charge screening of surface carboxyl groups.¹⁰⁸ By flow focusing CNF suspensions and then inducing gelation through pH reduction or salt addition, aligned hydrogels with oriented fibrils can be created and used as templates for anisotropic nanocomposites (Figure 6).^{111,112}

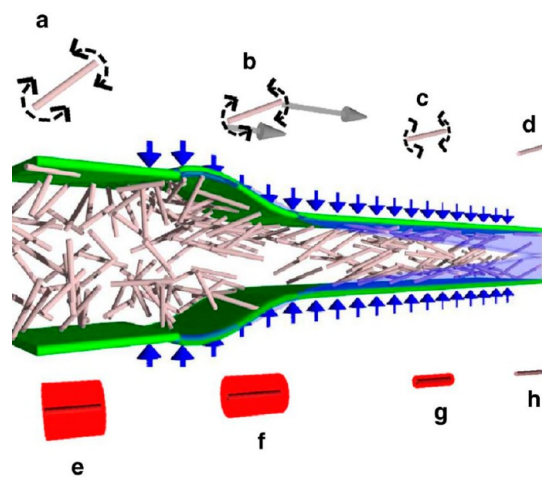


Figure 6. Illustration of the flow focusing CNF hydrogel assembly process. The diffusion of Na^+ ions into the flow channel is depicted by blue arrows, the concentration of which increases as the channel narrows, shown by the shaded blue area. (a) Initially, CNFs undergo Brownian diffusion, which decreases as (b) individual particles undergo hydrodynamically induced alignment. (c) Particles are constrained uniaxially, and (d) “frozen” in place as the increased Na^+ ions cause gelation of the system. (e–h) The electrostatic repulsion and the Debye length of CNF fibrils (shown in red) decrease along the channel as surface carboxyl groups become uncharged/shielded. Reprinted with permission from ref 112, copyright 2014 Nature Publishing Group.

Lundahl et al. also demonstrated flow-induced CNF alignment by wet spinning in order to create aligned filaments.¹¹³ The oriented filaments displayed exceptional mechanical properties that were enhanced as the degree of orientation was increased; however, filament stability drastically decreased in the presence of water, representing a key limitation of this technique.¹¹³

While most CNFs contain both amorphous cellulose and crystalline cellulose I (the native crystal structure of cellulose), Abe and Yano explored the effects of alkaline pretreatment on pulp before defibrillation as a way to alter the crystal structure. Increasing the NaOH concentration led to an increase in the cellulose II allomorph, attributed to axial shrinking of the CNFs in alkaline solution and, in effect, the preparation of cellulose II CNF hydrogels.^{114,115} Moreover, the hydrogels with the predominantly cellulose II crystal structure showed an increase in Young's modulus versus those with the cellulose I crystal structure, presumably due to the firm interdigitation of neighboring cellulose II nanofibers.¹¹⁵ While this work clearly highlights that pre- and postchemical modification can change the cellulose structure (and in turn the gel mechanical properties), the majority of works do not report of the crystal structure or degree of crystallinity of the CNF materials employed.

As with CNC reinforced hydrogels, CNF hydrogels are promising candidates for a wide range of tissue engineering and biomedical applications. Bhattacharya et al. demonstrated the applicability of homogenized, autoclaved CNF hydrogels as cell

culture scaffolds, showing minimal cytotoxicity and induced spheroid formation in HepaRG and HepG2 cells.¹¹⁶ However, these hydrogels displayed relatively weak mechanical properties, prompting several research groups to focus on designing stiffer gels through ion-mediated gelation.^{117–120} By using metal salts (Ca^{2+} , Fe^{3+}) to induce gelation, Zander et al. covalently attached fibronectin protein to CNF gels to increase fibroblast cell adhesion.¹²⁰ Masruchin et al. utilized negatively charged TEMPO-oxidized CNFs and mono-, di-, and trivalent cations to create gels,¹¹⁸ with higher porosity, physical cross-link density, mechanical strength, and swelling ratio observed at higher cross-linking ion valencies (attributed to stronger ionic cross-linking). Dong et al. identified similar effects following investigation of the effects of di- and trivalent cations on the network formation of carboxylated CNFs.¹¹⁷ Radioactive uranyl cations have a similar gelation effect on CNF suspensions and are adsorbed at concentrations up to 167 mg/g, leading to simultaneous gel formation, showing promise for the removal of radioactive metals from wastewater.¹²¹

Similar to most CNC hydrogels, CNF hydrogels lack macroscopic structure or hierarchy, which is serious disadvantage when developing biomimetic materials or hydrogels for many biomedical applications. Torres-Rendon et al. utilized sacrificial templating of a resin material in order to control the 3D structure of CNF hydrogels (Figure 7).¹²² A mixture of methacrylates and acrylamides with a hydrolytically labile cross-linker were lithographically 3D printed to form a reverse template, which can be filled with CNF via centrifugation and subsequently submerged in NaOH solution to dissolve the sacrificial template. The resulting macroporous hydrogel scaffolds supported growth of mouse fibroblast cells at a level comparable to tissue culture polystyrene and allowed for adherence and spreading of human mesenchymal stem cells.¹²² Finally, Lauren et al. demonstrated a method of radiolabeling CNFs with technetium-99m, allowing for the tracking/monitoring of the localization of CNF hydrogels and evaluation of drug delivery in vivo as a function of time,¹²³ further showing that CNF hydrogels are promising for a variety of tissue engineering applications.

2.5. Hydrogels with Physically and Chemically Entrapped CNFs. CNFs have been widely used as reinforcing agents/fillers to create tough yet highly flexible polymer gels, with much emphasis on specific biomedical applications (Table 3). As with CNC-based gels, the swelling ratio and mechanical properties of CNF nanocomposite hydrogels readily lend themselves to soft tissue applications, including nucleus pulposus replacement,^{124,125} and tendon/ligament/cartilage engineering.^{126–128} Common polymer matrices for CNF hydrogels are similar to those used for CNC-hydrogels including PVA, polyacrylamides and polyacrylates, PEG-based polymers, cellulose derivatives, chitosan, collagen, alginate, and gelatin. Hydrogel preparation is generally performed by straightforward mixing (homogenization), in situ polymerization, or heat-treatment to give hydrogels with up to 2.5 wt % CNFs (i.e., lower loadings than CNC gels, due to the propensity for entanglement of CNF fibers), higher swelling ratios, and compressive moduli on the order of 1000 kPa depending on polymer component, as indicated in Table 3. For example, networks of *N*-vinylpyrrolidone and CNFs were found to swell up to six times their original hydrated mass (depending on cross-link density) and were found to be noncytotoxic to fetal cartilage cells.^{124,125}

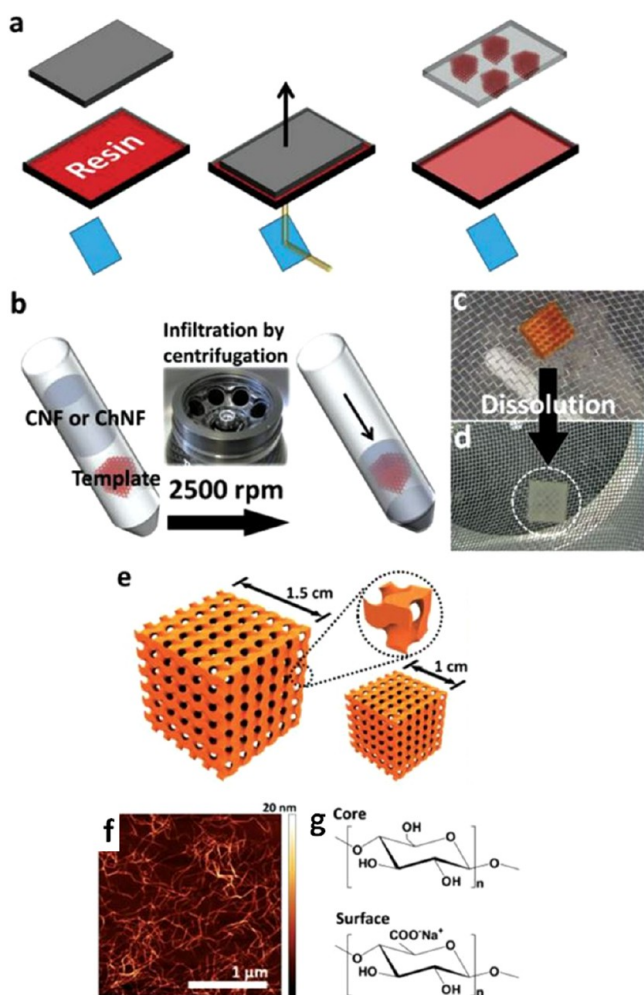


Figure 7. (a) Preparation of CNF template hydrogels via lithographic printing, (b) centrifugation of CNF into the template, and (c, d) dissolution of the sacrificial structure. (e) Typical scaffold structure and dimensions, (f) AFM images of the CNF nanofibrils, and (g) chemical structure within the gel are also shown. This technique presents a versatile method of creating structured hydrogels, which broadens the applications of these materials. Adapted with permission from ref 122, copyright 2015 John Wiley & Sons, Inc.

Given that many biological tissues exhibit mechanical anisotropy or gradients, efforts to introduce anisotropic gel structures in CNF-containing hydrogels have also been made. Wang et al. demonstrated additive manufacturing of CNF-based nanocomposite hydrogels with mechanical gradients via direct filament writing.¹²⁹ By decreasing the fraction of CNF to synthetic copolymer, substantial toughening of the hydrogel was achieved. These hydrogels may then be printed in a predefined compositional gradient, as shown in Figure 8, and allowed to heal to form a continuous film with strain gradients. Such a film is potentially useful as complex “origami actuators” or as tools to study material/cell interactions with CNFs.¹²⁹

Self-healing and drug delivery applications of CNF-polymer hydrogels have also been explored. Ionic interactions between amine-terminated PEG and carboxylated CNFs allow these nanocomposite hydrogels to exhibit a reversible sol–gel transition upon either strain or temperature ramping.¹³⁰ PNIPAm hydrogels have also been used to create CNF reinforced thermosensitive hydrogels.^{131,132} Wei et al. showed that these nanocomposite hydrogels exhibited characteristic

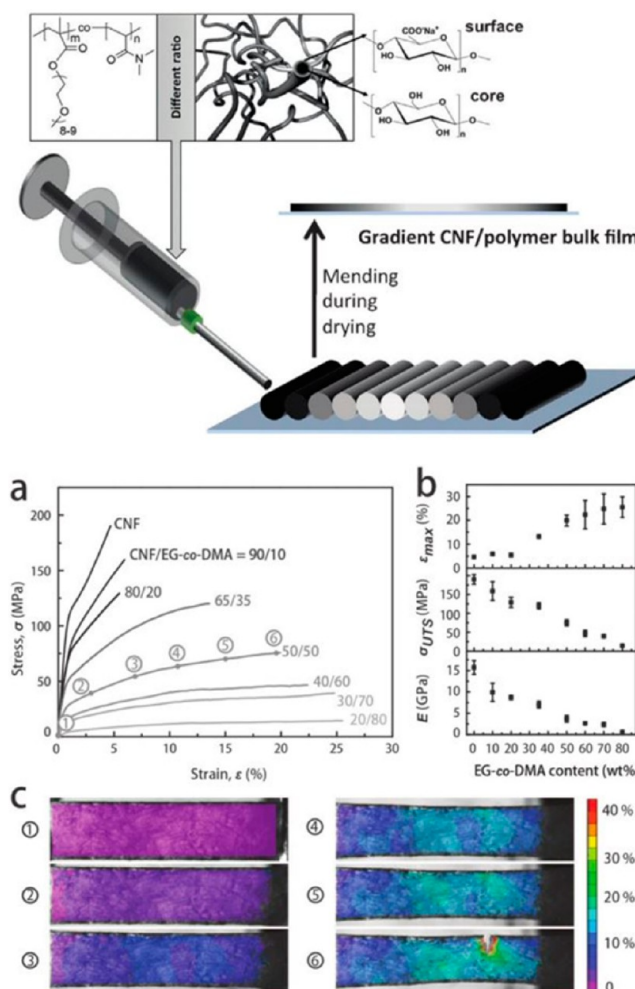


Figure 8. (top panel) Schematic of the additive manufacturing of gradient CNF/synthetic copolymer (ethylene glycol methyl ether methacrylate-co-dimethacrylamide, EG-co-DMA) hydrogels made via direct filament writing with varying weight ratios of starting components. Upon drying, the separate filaments undergo “healing” to form mechanically coherent bulk films with tunable mechanical gradients, enabling sophisticated modeling of biological systems or complex strain actuators. (a, b) Mechanical properties of the hydrogels and (c) strain fields during tensile testing of the nanocomposites are also shown. Adapted with permission from ref 129, copyright 2016 John Wiley & Sons, Inc.

temperature dependent swelling, which along with compressive modulus was enhanced with increasing CNF content.¹³¹ Alternately, Syverud et al. cross-linked CNFs with PEI and allylamine modified PNIPAm particles to create freeze–thaw structured thermosensitive gels.¹³²

CNF nanocomposite hydrogels have been used as absorbents due to their relatively high swelling capacity.^{133,134} Yue et al. created macroscopic core–shell hydrogels based on PVA and alginate that showed higher storage and shear moduli accompanied by slightly increased dye absorption capacity when CNFs were chemically cross-linked into the core.¹³⁵ In addition, Chinga-Carrasco et al. showed that diamine cross-linked CNF gels exhibited pH-responsive swelling due to the pK_a of surface carboxyl groups.¹³⁶ Overall, research has shown that CNFs are compatible with a wide variety of synthetic and natural polymers and preparation methods, with chemical

Table 4. List of Material Properties for “CNC-Only” Aerogels and Aerogels with Physically and Chemically Entrapped CNCs^a

Materials ^b	CNC Surface Chemistry	Preparation Method	Modulus (kPa)	Density (g/cm ³)	Porosity (%)	Specific Surface Area (m ² /g)	Post-Treatment	Intended Application	Ref
“CNC-Only” Aerogels									
CNC	Sulfate half-ester	CPD	–	0.078 – 0.155	91 – 95	216 – 605	–	–	149
CNC	Sulfate half-ester	Freeze dried	–	–	–	17 – 167	Gas-phase palmitoylation	–	154
CNC	COOH	Acid-gelation, CPD	≈ 40 – 900	0.004 – 0.04	98.1 – 99.7	≈ 500 – 600	–	Insulation	150
CNC	Aldehyde, hydrazide	Solvent exchange to EtOH, CPD	–	0.0056 – 0.0217	98.6 – 99.6	250	–	Absorbent	148
CNC	Sulfate half-ester	Freeze dried	–	–	–	91.5 – 93.9	–	–	158
CNC	COOH	CPD	–	0.0057 – 0.22	86.1 – 99.6	43 – 429	ALD	–	159
CNC	Sulfate half-ester	CPD	–	–	–	261 – 354	–	–	160
CNC	Acetylated	Freeze dried	–	0.0034	–	–	–	Oil absorbent	161
Physical Entrapment of CNCs into Aerogels									
Clay gel, PVA, CNC	Sulfate half-ester	Freeze dried	18 – 788	0.01 – 0.101	–	–	–	–	156
Whey protein, CNC	Sulfate half-ester	Freeze dried	≈ 530	0.11 – 0.16	–	1.72	–	Oil encapsulation	155
Clay gel, PVA, CNC	Sulfate half-ester	Freeze dried	≈ 10 – 75	0.055 – 0.108	81.6 – 84.9	–	–	–	157
PVA, SA, CNC, crosslinker	HCl hydrolyzed	Freeze dried	≈ 1600 – 3600	–	90.8 – 95.2	–	–	–	162
Chemical Entrapment of CNCs into Aerogels									
CNC, (“trapped” PPy, CNT, or MnO ₂)	Aldehyde, hydrazide	Freeze dried	–	0.017	98.9 – 99.3	–	–	Supercapacitor	151,152
CNC, CMC (“trapped” MOF)	Aldehyde, hydrazide	Freeze dried	–	–	–	–	–	Separations	153
POEGMA, CNC	Aldehyde	Freeze casting	–	–	–	5 – 33	–	Biomaterials	97

^aSection entries are ordered chronologically. The “–” indicates information was not provided in the publication. ^bPolymer acronyms used: CNT (carbon nanotubes), CMC (carboxymethylcellulose), MOF (metal organic frameworks), PPy (polypyrrole), PVA (poly(vinyl alcohol)), and SA (sodium alginate). CPD = critical point dried; ALD = atomic layer deposition.

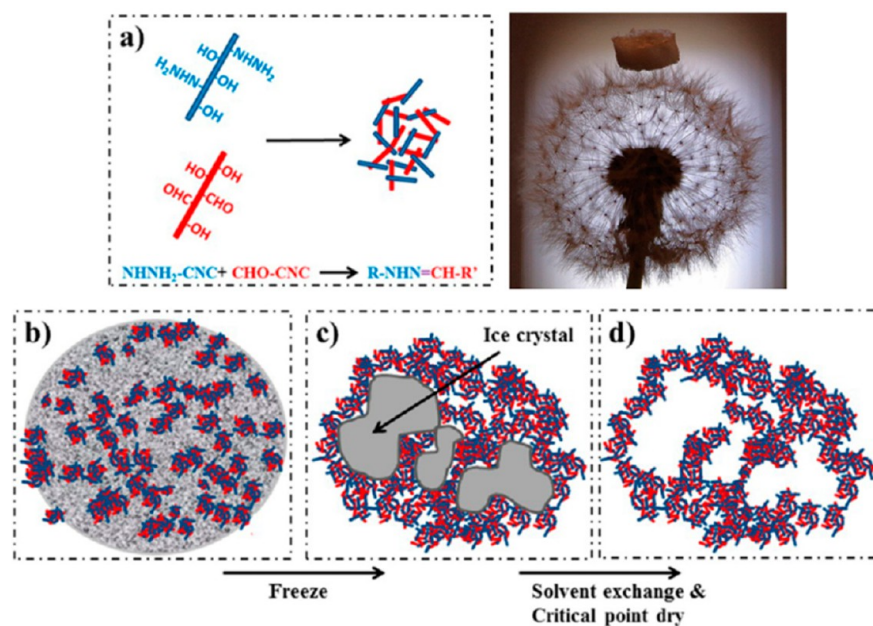


Figure 9. CNC aerogels from aldehyde and hydrazide-functionalized CNCs (red and blue rods). (a) Vortex mixing causes small “clusters” to form in suspension, (b) which are then frozen. (c) Ice crystal growth pushes the clusters closer together, (d) to form a free-standing structure upon ice sublimation. A lightweight aerogel is shown sitting on a dandelion in the top right panel. These aerogels exhibit a bimodal pore size distribution due to the cross-linking between individual CNCs and the ice crystal growth, leading to enhanced stability and mechanical performance along with high absorption capacity. Adapted from ref 148, copyright 2014 American Chemical Society.

cross-linking enhancing hydrogel physical properties to create

high performance materials across a wide range of applications.

3. AEROGELS

3.1. CNC Aerogels. CNC aerogels can approach the high porosity and surface area of silica-based aerogels but are much

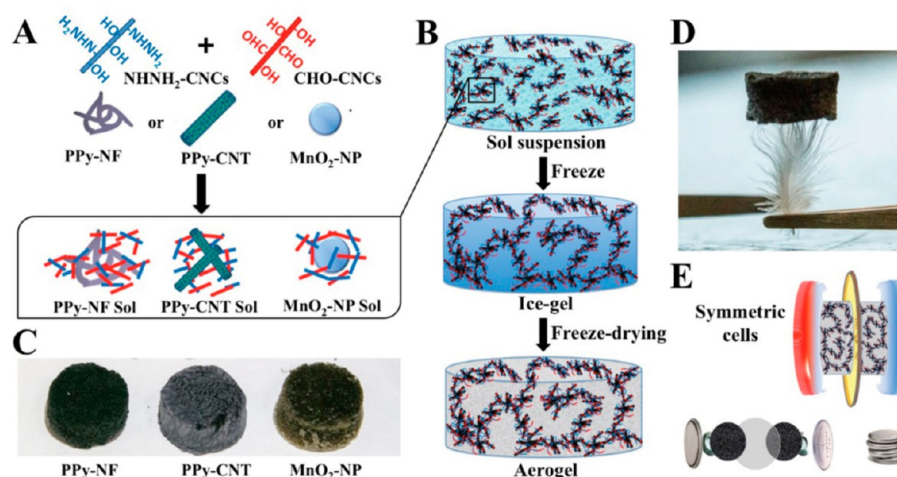


Figure 10. (A) Schematic representation of conductive CNC aerogel components, (B) the freeze-drying process, and (C) representative aerogels with a diameter of 1.5 cm, (D) that are light enough to rest on top of a feather. (E) A schematic setup of the supercapacitor cells using two aerogels separated by a polyethylene membrane. The extremely lightweight and flexible CNC aerogels allow for the incorporation of a high mass ratio of active capacitive nanoparticles, offering a large surface area to promote charge storage and numerous paths for diffusion to lower device internal resistance. Reprinted with permission from ref 151, copyright 2014 John Wiley & Sons.

less fragile, making them useful in applications requiring mechanical flexibility and strength.^{148,149} Table 4 summarizes the current literature on CNC aerogels, including “CNC-only” aerogels and those whereby CNCs reinforce a dry porous polymer network through physical or chemical cross-linking. The table describes the CNC surface chemistry, aerogel preparation method, physical properties (density, porosity, surface area), and intended application. Some mechanical properties are presented, but it is exceedingly difficult to accurately compare the mechanics between materials and test methods. In general, CNC aerogels are prepared by critical point drying or freeze-drying, sometimes with solvent exchange steps and postpreparation chemical modifications. The elastic modulus can be as high as hundreds of kPa, even for lightweight materials (0.0056 to 0.155 g/nm³) with porosities ranging from 82% to 99.6% (Table 4).

Structured aerogels have also been described by Kobayashi et al., who were able to induce gelation of TEMPO oxidized liquid crystalline cellulose nanofibers (possessing some amorphous cellulose and CNC-like properties) using a dilute acid solution, which could then undergo critical point drying to form aerogels with nematic structures.¹⁵⁰ These structured lightweight aerogels have a high transmittance in the visible-light region and are more similar to silica aerogels than to other cellulose aerogels in terms of compressive modulus and surface area. These aerogels also display low thermal conductivity and could therefore have application as high performance insulating materials.¹⁵⁰

Surface areas up to 605 m²/g have been reported in relatively delicate CNC aerogels;¹⁴⁹ chemical cross-linking provides a route to create more mechanically stable aerogels with shape recovery abilities without compromising hugely on surface area.¹⁴⁸ Yang et al. demonstrated that hydrazone cross-linked CNC aerogels (Figure 9) displayed greater than 85% shape recovery after 20 cycles of 80% compression in water. These aerogels may be useful in oil/water separation applications as they preferentially absorb and hold water (up to 160 g/g) in a mixture of water and dodecane.¹⁴⁸ This chemically cross-linked “CNC-only” aerogel platform furthermore served as flexible

freestanding universal supports for a number of functional nanoparticles as described below.^{151–153}

Preferential oil/water absorption or partitioning is a common application for CNC aerogels. As these structures are naturally hydrophilic, post-treatment of the aerogels is required to promote specific CNC–oil interactions. Fumagalli et al. subjected CNC aerogels to a gas-phase palmitoylation which led to the ability to redisperse the freeze-dried CNCs in nonpolar solvents (hence, disassembling the aerogel).¹⁵⁴ Alternatively, combining CNCs with another raw material can also lead to increased hydrophobicity or absorption capacities. For example, Ahmadi et al. developed CNC/whey protein aerogels for loading of hydrophobic nutraceuticals, showing a fish oil loading efficiency of 71.7%.¹⁵⁵

The inclusion of functional nanoparticles or polymers into CNC aerogels represents the rest of the examples shown in Table 4. By incorporating metal organic frameworks (MOFs) at up to 50 wt %, Zhu et al. were able to match or increase the adsorption capacity of the MOF powder on its own, demonstrating removal of wastewater contaminants such as heavy metals, carcinogenic dyes, and drug precursors.¹⁵³ The absorption and separation abilities are attributed to the hierarchical structure of the porous CNC aerogels allowing fast water uptake and swelling through macropores and capillary action of mesopores, facilitating interactions of the MOF micropores with the contaminants. Furthermore, the compression stability of the CNC aerogels allowed water to be squeezed out while retaining a high concentration of the contaminants.¹⁵³

Several other material properties of CNC aerogels may be enhanced through incorporation of nanoparticles, including strength, flexibility, and conductivity. Nanoclay/PVA-based aerogels have been shown to exhibit good mechanical properties, with CNCs contributing to increased modulus, crack prevention, and shape recovery.^{156,157} Meanwhile, loading conductive materials such as polypyrrole nanofibers, carbon nanotubes, or manganese dioxide nanoparticles leads to lightweight aerogels with excellent capacitance retention at high charge–discharge rates for supercapacitor devices (Figure 10).^{151,152} Such materials have shown up to 97.81% capacitance

Table 5. List of Material Properties for CNF-Only Aerogels and Aerogels with Physically and Chemically Entrapped CNFs^a

Materials ^b	CNF Surface Chemistry	Preparation Method	Modulus (kPa)	Density (g/cm ³)	Porosity (%)	Specific Surface Area (m ² /g)	Post-Treatment	Intended Application	Ref
"CNF-Only" Aerogels									
CNF	Unspecified, from Domsjö	Freeze dried	203 – 240	≈ 0.02	≈ 98	20 – 66	Polyaniline coated	Conductive materials	177
CNF	Carboxy-methylated	Freeze dried	–	0.00027 – 0.03	98.2 – 99.7	11 – 15	CVD	Hydrophobic materials	170
CNF	COOH or OH	Freeze dried	–	0.0002 – 0.017	≈ 99	–	–	Sound absorption	164,165
CNF	COOH	Freeze dried from t-butanol	35 – 2800	0.014 – 0.105	92.8 – 99	153 – 284	–	–	201
CNF	Unspecified, from UPM	Vacuum dried	–	0.02	98.6	–	CVD to fluorinate	Hydrophobic materials	172
CNF	Carboxy-methylated from Innventia	Freeze dried	–	0.03	98	20	CVD TiO ₂	Photo-switchable absorption	169
CNF	Unspecified, from UPM	Freeze dried or CPD	–	0.02 – 0.03	> 98	–	ALD, calcination or TiO ₂ coating	Encapsulation, absorbents	173,178
CNF	Carboxy-methylated	High speed mixing, freeze dried	–	0.004 – 0.014	99.1 – 99.8	11 – 42	CVD	Oil absorption	171
CNF	OH or COOH	Freeze dried	–	–	–	–	–	Drug delivery	202
CNF	COOH	Freeze dried	–	–	–	–	–	–	203
CNF	COOH	Freeze dried	–	–	> 98	–	–	Dye-sensitized solar cells	175
CNF	COOH	Freeze dried, cyclic freeze-thaw	54.5	0.0017 – 0.0187	98.8 – 99.9	12.72 – 117.8	CVD	Absorbents	167,168
CNF	Unspecified	Solvent exchange, ambient drying	6 GPa	≈ 0.6	–	208	–	Gas filtration	163
CNF	OH	Freeze dried	–	–	–	20.1	CVD	Hydrophobic materials	174
CNF	COOH	Freeze dried	–	–	–	–	LbL	Supercapacitor	179
CNF	COOH	Freeze dried	–	–	–	≈ 60 – 90	–	–	118
CNF	COOH	Freeze dried	13 – 176	≈ 0.01 – 0.15	–	116 – 162	–	Insulation	204
CNF	OH	Freeze dried or CPD	–	–	–	103 – 1873	Pyrolysis	Supercapacitor	176
CNF	COOH	Freeze casted	–	1.36 – 1.47	–	–	–	–	166
CNF	COOH	Spray freeze dried	< 5700	≈ 0.01 – 0.03	98 – 99	–	–	Insulation	198
CNF	COOH	Freeze dried	–	0.0034 – 0.0051	≈ 99.7	94.8 – 195.5	Silanization	Oil absorption	205
Physical Entrapment of CNFs into Aerogels									
Aminosilane, CNF	Unspecified, from Arbocel	Freeze dried	–	0.026 – 0.061	–	7.1 – 26.8	–	CO ₂ capture	194
PPy, CNF	COOH	Air dried, freeze dried or CPD	0.51 GPa	1.47	30 – 98	1 – 246	–	Conductive materials	181
HA, CNF	COOH	Freeze dried	75 – 200	0.008 – 0.011	99.3 – 99.5	13.8 – 19.8	–	–	199
CNF, Ag nanoparticles	COOH	Freeze dried	–	–	–	8 – 20	–	–	119
GO, CNF	COOH	CPD	–	–	–	–	–	Flexible supercapacitor	183
CNT, CNF	COOH	CPD	–	–	–	–	–	Flexible supercapacitor	184
Soy protein, CNF	Unspecified	Freeze dried	–	0.111 – 0.115	92.7	0.19 – 1.9	–	Sorption	188
TiO ₂ /SiO ₂ , CNF	COOH	Freeze dried	–	0.0089	> 99.5	6.1	Calcination	Absorption	187
CNT, CNF	OH	Freeze dried	410 – 528	0.01 – 0.024	98.4 – 99.7	–	–	Conductive materials	185
Montmorillonite clay, CNF	Unspecified	Freeze dried or casted	25.8 – 386 or 2130 – 3860	29.8 – 44.8 or 0.31 – 0.37	–	–	–	Thermally stable packaging	196
PMSQ, CNF	OH	Heat cured, CPD	–	0.02 – 0.186	–	525 – 732	–	Thermal insulation	195
TiO ₂ , CNF	OH	Freeze dried	–	–	–	–	Calcination or pyrolysis	Conductive materials	192
CNF, PVA	COOH	Freeze casted	30 – 10440	0.016 – 0.018	98.7 – 98.8	–	CVD	Hydrophobic materials	186
CNF, PNIPAm	COOH	Freeze dried	20 – 220	0.018 – 0.034	–	44.05 – 55.42	–	–	189
Chemical Entrapment of CNFs into Aerogels									
CNF, kymene	COOH	Freeze dried	–	–	–	–	–	–	206
Cross-linker, CNF	Carboxy-methylated	Freeze dried	–	–	≈ 98	–	LbL	Conductive materials	180
PVA, GO, CNF	COOH	Cross-linked, Freeze dried	4 – 100	0.015 – 0.035	–	140 – 243	–	Insulation	197
Collagen, CNF	Aldehyde	Freeze dried	–	0.0206 – 0.032	90.48 – 95.45	–	–	Wound dressing	200
CNF, MTMS	Unspecified	Freeze dried	27.7	0.00507 – 0.0173	99.0 – 99.7	3.1 – 25.0	–	Oil sorption	207
CNF, cross-linker resin	Unspecified, from University of Maine	Freeze dried	–	0.01 – 0.02	–	–	Carbonization	Oil sorption	208
CNF, PVA	COOH	Emulsification, freeze dried	–	0.0046 – 0.0165	98.7 – 99.7	43.8 – 117	CVD	Hydrophobic materials	190
CNF, crosslinker	COOH	Atomization-freeze dried	–	≈ 0.0021	–	–	–	Extraction/separation	193
CNF, MDI	COOH	Freeze dried	44 – 209	0.0069 – 0.0083	–	123 – 228	–	Oil/water separation	191

^aSection entries are ordered chronologically. The "–" indicates information was not provided in the publication. ^bPolymer acronyms used: CNT (carbon nanotube), GO (graphene oxide), HA (hydroxyl apatite), MDI (methylene diphenyl diisocyanate), MOF (metal organic framework), MTMS (methyltrimethoxysilane), PMSQ (polymethylsilsesquioxane), PNIPAm (poly *N*-isopropylacrylamide), PPy (polypyrrole), and PVA (poly(vinyl alcohol)). CPD = critical point dried, LbL = layer-by-layer deposition, CVD = chemical vapor deposition, and ALD = atomic layer deposition.

retention after 1000 cycles, extremely low internal resistance, and specific capacitance as high as 2.1 F/cm^2 , performance that is attributed to the high strength/flexibility and highly porous nature of the aerogels that provides good contact with the current collector and increased ionic paths for electrolyte diffusion.^{151,152} Although a relatively new field (at least with respect to CNF aerogels), CNC aerogels have proven highly versatile, targeting applications in absorption, separation, conductive templates, and biomaterials.

3.2. CNF Aerogels. Due to high chain entanglement, highly stable “CNF-only” aerogels are relatively easy to prepare via freeze-drying or critical point drying and have garnered much more research attention than their CNC counterparts. Table 5 shows 20 examples of “CNF-only” aerogels and another 20 in which CNFs are combined with other polymer networks or functional nanomaterials to form aerogels. In some cases, CNF-derived aerogel moduli reach as high as 6 GPa with densities as low as 0.0046 g/cm^3 (99.7% porous). While surface areas are generally lower than for CNC aerogels, there are a number of examples in which the aerogels are carbonized/calculated to achieve extremely high surface areas (ca. $1900 \text{ m}^2/\text{g}$). There is also more variety in postpreparation surface modification that has been explored with CNF aerogels, expanding the potential uses of these materials (Table 5).

Techniques other than freeze-drying or critical point drying have also been explored for the creation of CNF aerogels. Interestingly, Toivonen et al. developed a method of CNF vacuum filtration followed by solvent exchange and ambient drying that produced aerogel membranes without the need for freeze-drying or critical point drying.¹⁶³ Unsurprisingly and regardless of preparation method, increasing the CNF content within the aerogels led to a collapsed sheet-like morphology with higher bulk density and lower surface area than most reported aerogels.^{164,165} Freeze casting has also been shown to be an easy way to impart structure or anisotropic morphologies to CNF aerogels, creating uniaxially aligned fibrils in the direction of ice crystal growth. However, aggregation due to pH reduction or salt addition hampers this orientation, potentially limiting the functionality of these structured CNF-only aerogels.¹⁶⁶

One limitation of native cellulose-based aerogels is that they have a tendency to disassemble and redisperse in water, mandating the use of chemical cross-linking or post-treatments including chemical vapor deposition (CVD) or other coating methods to yield stable aerogels capable of high liquid absorption.^{167–169} In particular, the hydrophobization of CNF aerogels has attracted much research attention for their potential in oil/water separations and organic contaminant entrapment.^{170–174} Jiang and Hsieh demonstrated the potential amphiphilic nature of CNF aerogels, which could absorb 210 g/g of water and 375 g/g of chloroform.¹⁶⁷ A subsequent silane modification rendered the aerogels hydrophobic/oleophilic, capable of adsorbing $139\text{--}356 \text{ g/g}$ nonpolar hydrocarbons, polar aprotic solvents, and oils.¹⁶⁷ Kettunen et al. pursued similar superabsorbent hydrophilic and hydrophobic aerogels by incorporating a thin film of titanium dioxide through CVD, resulting in water absorbent/repellent photoswitchable aerogels.¹⁶⁹ Coated aerogels displayed a water absorbent capacity of 0.6 g/g , which increased substantially to 16 g/g upon UV illumination; the water repellent properties were subsequently recovered after storing the aerogels in the dark.

Postpreparation surface modifications of CNF aerogels have also been performed to create functional templates for

conductive applications. The impressive mechanical strength, high porosity, and flexibility of CNF aerogels lends these materials to solid supports for liquid electrolytes in dye-sensitized solar cells¹⁷⁵ and supercapacitor materials.¹⁷⁶ Miettunen et al. demonstrated that CNF-based electrolyte templates did not affect the efficiency, resistance, or stability of traditional liquid electrolyte solar cells; furthermore, Pääkkö et al. showed that dip coating the aerogels in conductive polymer led to the formation of conductive networks.¹⁷⁷ By performing pyrolysis and subsequent CO_2 activation, Zu et al. synthesized extremely high surface area carbon aerogels (up to $1873 \text{ m}^2/\text{g}$, 6-fold greater than the as-prepared CNF aerogel template) with excellent electrochemical performance for supercapacitor applications.¹⁷⁶ Finally, Korhonen et al. performed atomic layer deposition to deposit inorganic oxides on CNF aerogels and subsequently calcinated the structures to create hollow inorganic nanotube aerogels with significant potential as functional materials for drug encapsulation, catalysis, filtration, or microfluidic applications (Figure 11).¹⁷⁸

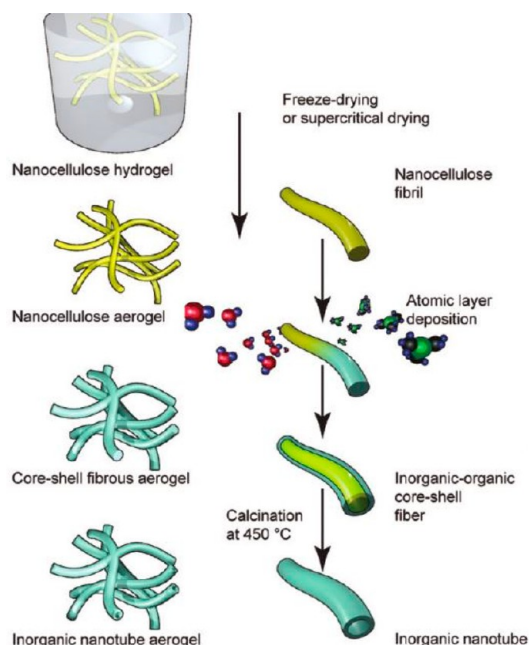


Figure 11. Schematic representation of the preparation of inorganic hollow nanotube aerogels from freeze-dried CNF aerogels. Physically entangled CNF hydrogels are first subjected to either freeze-drying or critical point drying to form aerogels. Subsequent atomic layer deposition of zinc oxide, titanium dioxide, or aluminum oxide led to uniform inorganic layers on the aerogel fibers. Calcination at 450°C easily degraded the cellulose cores, leaving hollow inorganic nanotube networks. This process is depicted for an entangled CNF aerogel (left) and for an individual CNF fibril (right). Reprinted from ref 178. Copyright 2011 American Chemical Society.

In a series of work by Nyström and co-workers, CNF aerogels were prepared following various drying methods and modified postpreparation using in situ polymerization or layer-by-layer (LbL) deposition to impart conductivity.^{179–182} Porosity, density, and conductivity greatly affected the voltammetric and chronoamperometric behavior of the CNF-based materials.¹⁸¹ LbL surface modification performed by flowing solutions of oppositely charged conductive polymers, biomolecules, and carbon nanotubes was demonstrated to enhance/initiate material properties such as compressive

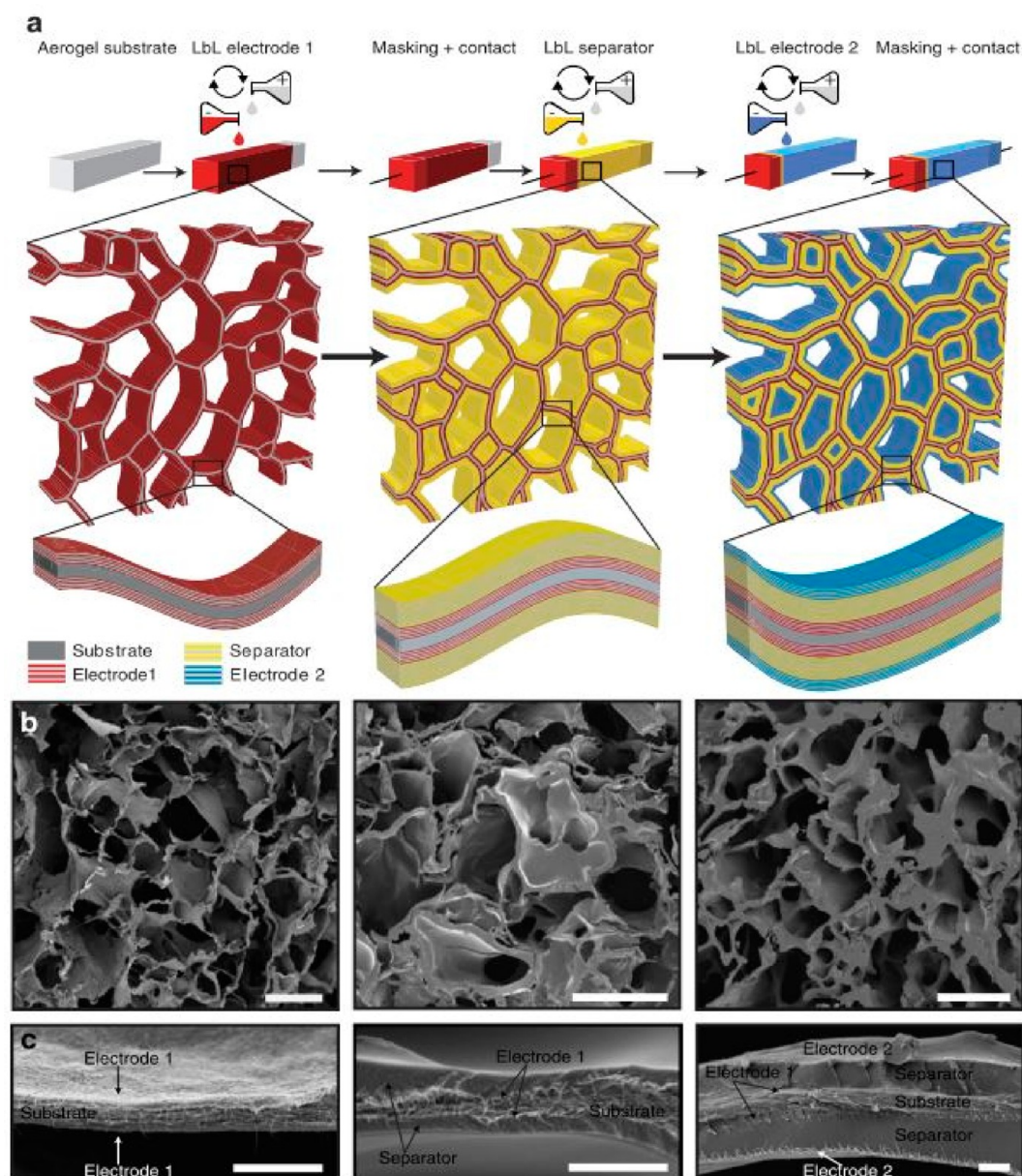


Figure 12. CNF energy storage device assembly and architecture. (a) Schematics of the LbL process on cross-linked CNF aerogels (gray “substrate”) to create electrode layers (5× PEI/CNT bilayers, shown in red and blue) with an insulating separation layer (30× PEI/PAA bilayers, shown in yellow) between. (b, scale bar 50 μm ; c, scale bar 2 μm) Cross-sectional SEM images of the device buildup are also shown. Reprinted with permission from ref 179, copyright 2015 Nature Publishing Group.

strength, fluorescence, mechanoresponsive resistance, and high charge storage capacity.¹⁸⁰ One of the most sophisticated devices demonstrated to date is an interdigitated 3D energy storage device, shown in Figure 12, that uses cationic polyethylenimine (PEI) and anionic carboxylated carbon nanotube layers as the electrodes and PEI/poly(acrylic acid) (PAA) as the insulating spacers.¹⁷⁹ These battery and supercapacitor CNF devices were operable under cyclic bending and compression without any observable structural damage and minimal changes in performance/charge capacity versus unstrained controls.¹⁷⁹

Gao et al. have also reported on the development of CNF-based solid-state flexible supercapacitors via the incorporation of graphene or multiwalled carbon nanotubes into the aerogels with the CNF structures acting as “spacers”, with increasing porosity leading to increased capacitance.^{183,184} The nominally

brittle carbon nanotube aerogels are increasingly ductile with CNF incorporation, promising for applications requiring flexibility including potentially “wearable” electrochemical applications.¹⁸⁵

Some researchers have chosen to directly prepare nanocomposite CNF-based aerogels both to remove the requirements for multistep postprocessing and to efficiently add functionality throughout the network. Both hydrophilic/hydrophobic absorbents^{186–191} and conductive aerogels^{181,183–185,192} are common applications for these materials. For example, Zhang et al. incorporated thermoresponsive PNIPAm into CNF aerogels and demonstrated a drastic temperature dependent water affinity in the materials; the aerogels floated and were nonwetting in hot water but sank and wetted once the water temperature was lowered.¹⁸⁹ Zhai et al. freeze-dried an emulsion of CNF and PVA followed by

subsequent cross-linking and silanization to create super-hydrophobic aerogel microspheres that were very effective as oil/organic solvent absorbents, with the absorption capacity increasing with increasing microsphere size.¹⁹⁰ More recently, Zhang et al. prepared ultralightweight CNF-based aerogel microspheres using ultrasonication for use as suspension microreactors in tailored separation/extraction applications.¹⁹³ The use of a covalent cross-linker drastically increased the aerogel stability in harsh environments and facilitated subsequent reusability without decreasing removal efficiency.¹⁹³ Other demonstrated applications of CNF nanocomposite aerogels include CO₂ capture,¹⁹⁴ thermal insulating materials,^{195–198} and biomedical applications, as discussed thoroughly for nanocellulose hydrogels.^{199,200} Notably, the ease of preparation and tailorability of CNFs has led to a drastic increase in their use in functional aerogels.

4. SUMMARY AND OUTLOOK

Interest in nanocellulose materials is growing rapidly; more than 50% of the references cited in this Review are post-2013. The research that has been conducted clearly indicates that the use of both CNCs and CNFs in hydrogels and aerogels leads to functional materials with many desirable properties suitable for a wide variety of applications across all fields of science and engineering. The recently implemented large scale commercial production of CNCs and CNFs, standardization, purification and characterization protocols, increasingly positive outlook on the toxicity profile, and the establishment of greener, more industrially feasible, and more reproducible nanocellulose surface modifications and processing methods all suggest that the use of CNCs and CNFs in applications ranging from small-scale medical-grade products through to larger-scale sorbent products is feasible.

Moving forward, there are still some challenges remaining before the commercialization of nanocellulose-based hydrogels and aerogels can be realized. First and foremost, an improved understanding of how the natural variability of nanocellulose as a function of the cellulose feedstock and production technique affects the resulting hydrogel/aerogel properties is required. As with all natural products, batch-to-batch variability will always exist and will be critical to monitor, particularly as the number of nanocellulose producers grows. There have been a few studies investigating the effects of nanocellulose aspect ratio, surface charge, or hydrolysis conditions on their resulting properties;^{16,62,209} however, systematic studies of these effects on the resulting properties of nanocomposites are largely lacking. Realistically, for commodity applications, the costs to produce the types of hydrogels and aerogels described in this Review are currently prohibitively high. Nevertheless, the combination of information presented in Tables 1–5 should help to identify the best nanocellulose type, chemistry, and method to obtain low cost materials with targeted properties. Furthermore, engineers continue to explore new large scale drying methods such as spray drying, freeze-drying, freeze-spray drying, and pressurized gas-expanded drying that may significantly improve the scalability and thus reduce the cost of aerogel production.^{198,210–212}

In many of the potential biomedical applications of these materials, costs are less constraining. However, thorough toxicity testing of all materials (before and after surface modifications and processing) remains essential and, to-date, is not widely executed in the published literature. Specifically, detailed in vivo and long-term biocompatibility studies to better

characterize the interaction between nanocellulose and biological tissue remain largely unexplored, with the available literature difficult to compare due to differences in nanocellulose properties as well as the type of animal model used (i.e., type of exposure, species, dose, and duration). As much as possible in the context of the different biomedical applications of nanocellulose hydrogels/aerogels, standardization of these assay components to glean a more accurate understanding of the effects of CNC/CNF surface modification and cross-linking on toxicity would be beneficial. Comprehensive toxicity screening for the use of nanocellulose-based composites in food and cosmetic applications, both of which require regulatory approval, is also still required and would similarly benefit from standardized testing protocols.

Improvements in specific targeted properties of CNC/CNF hydrogels and aerogels would also enhance the potential of such nanocomposites in applications and suggest promise for future research in this area. Particularly for biomedical applications, the fabrication of hydrogels that can be administered in a minimally invasive manner (i.e., by injection) but feature controlled anisotropic properties governed by CNC/CNF orientation would be highly beneficial for better mimicking native tissues with anisotropic alignments and mechanics (e.g., muscle). Interpenetrating/double network hydrogels in which CNCs represent the stiff phase may offer highly impactful increases in the modulus and/or extensibility of hydrogels versus current materials, with CNC/CNF alignment adding the potential for anisotropic responses in relevant applications (e.g., strain sensors). Further enhancements in the mechanical strength of nanocellulose-based aerogels would significantly improve the capacity of such nanocomposites for high-pressure and/or long-term use, as current materials remain relatively soft (particularly as the porosity and thus desirable surface area increase); the development of nanocellulose-compatible methods for fabricating controllable multiscale pore structures may be highly useful in this context. Finally, significant opportunities exist for manipulating the high surface area of CNCs (via simple chemistry relative to comparable inorganic fillers) to drive enhancements in specific adsorption capacity, increased resolution of sensing events, controlled delivery of bound therapeutics or growth factors, and/or interfacial conductivity to improve the performance of nanocellulose hydrogel/aerogel composites in a range of target applications.

Overall, nanocellulose's high strength, high aspect ratio, commercial availability, and potential for chemical modification have opened a number of new application areas in selective absorption, insulation, functional templates, and biomedicine (among others). The literature has shown that, in general, short rigid CNCs serve to effectively reinforce composite materials, adding toughness and flexibility, and template aligned structures, while longer elastic CNFs act as flexible gels with the option of facile postmodification. Furthermore, the anisotropic nature of nanocellulose has led to several unique and effective methods of creating structured materials across several length scales. We anticipate that ongoing research will further expand the scope of both performance and applications achievable with CNC- and CNF-based hydrogels and aerogels leading to both high-value specialty materials and high-volume commodity materials.

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REFERENCES

- (1) Moon, R. J.; Martini, A.; Nairn, J.; Simonsen, J.; Youngblood, J. Cellulose Nanomaterials Review: Structure, Properties and Nanocomposites. *Chem. Soc. Rev.* **2011**, *40* (7), 3941–3994.
- (2) Habibi, Y.; Lucia, L. A.; Rojas, O. J. Cellulose Nanocrystals: Chemistry, Self-Assembly, and Applications. *Chem. Rev.* **2010**, *110* (6), 3479–3500.
- (3) Klemm, D.; Kramer, F.; Moritz, S.; Lindström, T.; Ankerfors, M.; Gray, D.; Dorris, A. Nanocelluloses: A New Family of Nature-Based Materials. *Angew. Chem., Int. Ed.* **2011**, *50* (24), 5438–5466.
- (4) Plackett, D. V.; Letchford, K.; Jackson, J. K.; Burt, H. M. A Review of Nanocellulose as a Novel Vehicle for Drug Delivery. *Nord. Pulp Pap. Res. J.* **2014**, *29* (1), 105–118.
- (5) Jorfi, M.; Foster, E. J. Recent Advances in Nanocellulose for Biomedical Applications. *J. Appl. Polym. Sci.* **2015**, *132*, 41719–41737.
- (6) Habibi, Y. Key Advances in the Chemical Modification of Nanocelluloses. *Chem. Soc. Rev.* **2014**, *43* (5), 1519–1542.
- (7) Lin, N.; Dufresne, A. Nanocellulose in Biomedicine: Current Status and Future Prospect. *Eur. Polym. J.* **2014**, *59*, 302–325.
- (8) Cowie, J.; Bilek, E. M.; Wegner, T. H.; Shatkin, J. A. Market Projections of Cellulose Nanomaterial-Enabled Products — Part 2: Volume Estimates. *TAPPI J.* **2014**, *13* (6), 57–69.
- (9) Mariano, M.; El Kissi, N.; Dufresne, A. Cellulose Nanocrystals and Related Nanocomposites: Review of Some Properties and Challenges. *J. Polym. Sci., Part B: Polym. Phys.* **2014**, *52* (12), 791–806.
- (10) Shatkin, J. A.; Wegner, T. H.; Bilek, E. M.; Cowie, J. Market Projections of Cellulose Nanomaterial-Enabled Products — Part 1: Applications. *TAPPI J.* **2014**, *13* (5), 9–16.
- (11) Roman, M. Toxicity of Cellulose Nanocrystals: A Review. *Ind. Biotechnol.* **2015**, *11* (1), 25–33.
- (12) Moon, R. J.; Schueneman, G. T.; Simonsen, J. Overview of Cellulose Nanomaterials, Their Capabilities and Applications. *JOM* **2016**, *68* (9), 2383–2394.
- (13) Domingues, R. M. A.; Gomes, M. E.; Reis, R. L. The Potential of Cellulose Nanocrystals in Tissue Engineering Strategies. *Biomacromolecules* **2014**, *15* (7), 2327–2346.
- (14) Dugan, J. M.; Gough, J. E.; Eichhorn, S. J. Bacterial Cellulose Scaffolds and Cellulose Nanowhiskers for Tissue Engineering. *Nanomedicine* **2013**, *8* (2), 287–298.
- (15) Beck-Candanedo, S.; Roman, M.; Gray, D. G. Effect of Reaction Conditions on the Properties and Behavior of Wood Cellulose Nanocrystal Suspensions. *Biomacromolecules* **2005**, *6* (2), 1048–1054.
- (16) Sacui, I. A.; Nieuwendael, R. C.; Burnett, D. J.; Stranick, S. J.; Jorfi, M.; Weder, C.; Foster, E. J.; Olsson, R. T.; Gilman, J. W. Comparison of the Properties of Cellulose Nanocrystals and Cellulose Nanofibrils Isolated from Bacteria, Tunicate, and Wood Processed Using Acid, Enzymatic, Mechanical, and Oxidative Methods. *ACS Appl. Mater. Interfaces* **2014**, *6* (9), 6127–6138.
- (17) Yu, H.-Y.; Zhang, D.-Z.; Lu, F.-F.; Yao, J. New Approach for Single-Step Extraction of Carboxylated Cellulose Nanocrystals for Their Use as Adsorbents and Flocculants. *ACS Sustainable Chem. Eng.* **2016**, *4* (5), 2632–2643.
- (18) Chen, L.; Zhu, J. Y.; Baez, C.; Kitin, P.; Elder, T. Highly Thermal-Stable and Functional Cellulose Nanocrystals and Nanofibrils Produced Using Fully Recyclable Organic Acids. *Green Chem.* **2016**, *18*, 3835–3843.
- (19) Leung, A. C. W.; Hrapovic, S.; Lam, E.; Liu, Y.; Male, K. B.; Mahmoud, K. A.; Luong, J. H. T. Characteristics and Properties of Carboxylated Cellulose Nanocrystals Prepared from a Novel One-Step Procedure. *Small* **2011**, *7* (3), 302–305.
- (20) Camarero Espinosa, S.; Kuhnt, T.; Foster, E. J.; Weder, C. Isolation of Thermally Stable Cellulose Nanocrystals by Phosphoric Acid Hydrolysis. *Biomacromolecules* **2013**, *14* (4), 1223–1230.
- (21) Araki, J.; Wada, M.; Kuga, S. Steric Stabilization of a Cellulose Microcrystal Suspension by Poly(ethylene Glycol) Grafting. *Langmuir* **2001**, *17* (1), 21–27.
- (22) Saito, T.; Kimura, S.; Nishiyama, Y.; Isogai, A. Cellulose Nanofibers Prepared by TEMPO-Mediated Oxidation of Native Cellulose. *Biomacromolecules* **2007**, *8* (8), 2485–2491.
- (23) Abitbol, T.; Cranston, E. D. Directed Assembly of Oriented Cellulose Nanocrystal Films. In *Handbook of Green Materials*; Oksman, K., Rojas, O., Quintus, P., Eds.; World Scientific: Hackensack, NJ, 2014; Vol. 3, pp 79–103.
- (24) Abitbol, T.; Cranston, E. D. Chiral Nematic Self-Assembly of Cellulose Nanocrystals in Suspensions and Solid Films. In *Handbook of Green Materials*; Oksman, K., Rojas, O., Quintus, P., Eds.; World Scientific: Hackensack, NJ, 2014; Vol. 3, pp 37–56.
- (25) De France, K. J.; Yager, K. G.; Hoare, T.; Cranston, E. D. Cooperative Ordering and Kinetics of Cellulose Nanocrystal Alignment in a Magnetic Field. *Langmuir* **2016**, *32* (30), 7564–7571.
- (26) Revol, J.-F.; Godbout, L.; Dong, X.-M.; Gray, D. G.; Chanzy, H.; Maret, G. Chiral Nematic Suspensions of Cellulose Crystallites; Phase Separation and Magnetic Field Orientation. *Liq. Cryst.* **1994**, *16* (1), 127–134.
- (27) Hu, Z.; Cranston, E. D.; Ng, R.; Pelton, R. Tuning Cellulose Nanocrystal Gelation with Polysaccharides and Surfactants. *Langmuir* **2014**, *30* (10), 2684–2692.
- (28) Pääkkö, M.; Ankerfors, M.; Kosonen, H.; Nykanen, A.; Ahola, S.; Osterberg, M.; Ruokolainen, J.; Laine, J.; Larsson, P. T.; Ikkala, O.; et al. Enzymatic Hydrolysis Combined with Mechanical Shearing and High-Pressure Homogenization for Nanoscale Cellulose Fibrils and Strong Gels. *Biomacromolecules* **2007**, *8* (6), 1934–1941.
- (29) Wågberg, L.; Decher, G.; Norgren, M.; Lindström, T.; Ankerfors, M.; Axnäs, K. The Build-up of Polyelectrolyte Multilayers of Microfibrillated Cellulose and Cationic Polyelectrolytes. *Langmuir* **2008**, *24* (3), 784–795.
- (30) Aulin, C.; Ahola, S.; Josefsson, P.; Nishino, T.; Hirose, Y.; Österberg, M.; Wågberg, L. Nanoscale Cellulose Films with Different Crystallinities and Mesosstructures—Their Surface Properties and Interaction with Water. *Langmuir* **2009**, *25* (13), 7675–7685.
- (31) Hoare, T. R.; Kohane, D. S. Hydrogels in Drug Delivery: Progress and Challenges. *Polymer* **2008**, *49* (8), 1993–2007.
- (32) Haque, M. A.; Kurokawa, T.; Gong, J. P. Super Tough Double Network Hydrogels and Their Application as Biomaterials. *Polymer* **2012**, *53* (9), 1805–1822.
- (33) Camarero-Espinosa, S.; Endes, C.; Mueller, S.; Petri-Fink, A.; Rothen-Rutishauser, B.; Weder, C.; Clift, M.; Foster, E. Elucidating the Potential Biological Impact of Cellulose Nanocrystals. *Fibers* **2016**, *4* (3), 21.
- (34) Catalán, J.; Ilves, M.; Järventaus, H.; Hannukainen, K.; Kontturi, E.; Vanhala, E.; Alenius, H.; Savolainen, K. M.; Norppa, H. Genotoxic and Immunotoxic Effects of Cellulose Nanocrystals In Vitro. *Environ. Mol. Mutagen.* **2015**, *56*, 171–182.
- (35) Catalán, J.; Rydman, E.; Aimonen, K.; Hannukainen, K.-S.; Suhonen, S.; Vanhala, E.; Moreno, C.; Meyer, V.; da Silva Perez, D.; Sneek, A. Genotoxic and Inflammatory Effects of Nanofibrillated Cellulose in Murine Lungs. *Mutagenesis* **2017**, *32*, 23–31.
- (36) Endes, C.; Mueller, S.; Kinnear, C.; Vanhecke, D.; Foster, E. J.; Petri-Fink, A.; Weder, C.; Clift, M. J. D.; Rothen-Rutishauser, B. Fate

of Cellulose Nanocrystal Aerosols Deposited on the Lung Cell Surface In Vitro. *Biomacromolecules* **2015**, *16*, 1267–1275.

(37) Gaharwar, A. K.; Peppas, N. A.; Khademhosseini, A. Nanocomposite Hydrogels for Biomedical Applications. *Biotechnol. Bioeng.* **2014**, *111* (3), 441–453.

(38) Hoffman, A. S. Hydrogels for Biomedical Applications. *Adv. Drug Delivery Rev.* **2002**, *54* (1), 3–12.

(39) Akimov, Y. K. Fields of Application of Aerogels (Review). *Instrum. Exp. Tech.* **2003**, *46* (3), 287–299.

(40) Ureña-Benavides, E. E.; Ao, G.; Davis, V. A.; Kitchens, C. L. Rheology and Phase Behavior of Lyotropic Cellulose Nanocrystal Suspensions. *Macromolecules* **2011**, *44* (22), 8990–8998.

(41) Way, A. E.; Hsu, L.; Shanmuganathan, K.; Weder, C.; Rowan, S. J. pH-Responsive Cellulose Nanocrystal Gels and Nanocomposites. *ACS Macro Lett.* **2012**, *1*, 1001–1006.

(42) Chau, M.; Sriskandha, S. E.; Pichugin, D.; Thérien-Aubin, H.; Nykypanchuk, D.; Chauve, G.; Methot, M.; Bouchard, J.; Gang, O.; Kumacheva, E. Ion-Mediated Gelation of Aqueous Suspensions of Cellulose Nanocrystals. *Biomacromolecules* **2015**, *16*, 2455–2462.

(43) Beck-Candanedo, S.; Viet, D.; Gray, D. G. Triphase Equilibria in Cellulose Nanocrystal Suspensions Containing Neutral and Charged Macromolecules. *Macromolecules* **2007**, *40* (9), 3429–3436.

(44) Boluk, Y.; Zhao, L.; Incani, V. Dispersions of Nanocrystalline Cellulose in Aqueous Polymer Solutions: Structure Formation of Colloidal Rods. *Langmuir* **2012**, *28* (14), 6114–6123.

(45) Lewis, L.; Derakhshandeh, M.; Hatzikiriakos, S. G.; Hamad, W. Y.; MacLachlan, M. J. Hydrothermal Gelation of Aqueous Cellulose Nanocrystal Suspensions. *Biomacromolecules* **2016**, *17* (8), 2747–2754.

(46) Dorris, A.; Gray, D. G. Gelation of Cellulose Nanocrystal Suspensions in Glycerol. *Cellulose* **2012**, *19* (3), 687–694.

(47) Gonzalez, J. S.; Ludueña, L. N.; Ponce, A.; Alvarez, V. A. Poly(vinyl Alcohol)/cellulose Nanowhiskers Nanocomposite Hydrogels for Potential Wound Dressings. *Mater. Sci. Eng., C* **2014**, *34* (1), 54–61.

(48) Abitbol, T.; Johnstone, T.; Quinn, T. M.; Gray, D. G. Reinforcement with Cellulose Nanocrystals of Poly(vinyl Alcohol) Hydrogels Prepared by Cyclic Freezing and Thawing. *Soft Matter* **2011**, *7* (6), 2373–2379.

(49) Han, J.; Lei, T.; Wu, Q. Facile Preparation of Mouldable Polyvinyl Alcohol-Borax Hydrogels Reinforced by Well-Dispersed Cellulose Nanoparticles: Physical, Viscoelastic and Mechanical Properties. *Cellulose* **2013**, *20*, 2947–2958.

(50) McKee, J. R.; Appel, E. A.; Seitsonen, J.; Kontturi, E.; Scherman, O. A.; Ikkala, O. Healable, Stable and Stiff Hydrogels: Combining Conflicting Properties Using Dynamic and Selective Three-Component Recognition with Reinforcing Cellulose Nanorods. *Adv. Funct. Mater.* **2014**, *24* (18), 2706–2713.

(51) Aouada, F. A.; de Moura, R.; Orts, W. J.; Mattoso, L. H. C. Preparation and Characterization of Novel Micro- and Nanocomposite Hydrogels Containing Cellulosic Fibrils. *J. Agric. Food Chem.* **2011**, *59*, 9433–9442.

(52) Bajpai, S. K.; Pathak, V.; Soni, B.; Mohan, Y. M. CNWs Loaded poly(SA) Hydrogels: Effect of High Concentration of CNWs on Water Uptake and Mechanical Properties. *Carbohydr. Polym.* **2014**, *106*, 351–358.

(53) Bajpai, S. K.; Pathak, V.; Chand, N.; Soni, B. Cellulose Nano Whiskers (CNWs) Loaded-Poly(sodium Acrylate) Hydrogels. Part-I. Effect of Low Concentration of CNWs on Water Uptake. *J. Macromol. Sci., Part A: Pure Appl. Chem.* **2013**, *50* (5), 466–477.

(54) Chen, Y.; Xu, W.; Liu, W.; Zeng, G. Responsiveness, Swelling, and Mechanical Properties of PNIPAA Nanocomposite Hydrogels Reinforced by Nanocellulose. *J. Mater. Res.* **2015**, *30* (11), 1797–1807.

(55) Karaaslan, M. A.; Tshabalala, M. A.; Yelle, D. J.; Buschle-Diller, G. Nanoreinforced Biocompatible Hydrogels from Wood Hemicelluloses and Cellulose Whiskers. *Carbohydr. Polym.* **2011**, *86* (1), 192–201.

(56) Larsson, E.; Boujemaoui, A.; Malmström, E.; Carlmark, A. Thermoresponsive Cryogels Reinforced with Cellulose Nanocrystals. *RSC Adv.* **2015**, *5* (95), 77643–77650.

(57) De France, K. J.; Chan, K. J. W.; Cranston, E. D.; Hoare, T. Enhanced Mechanical Properties in Cellulose Nanocrystal-Poly(oligo Ethylene Glycol Methacrylate) Injectable Nanocomposite Hydrogels through Control of Physical and Chemical Cross-Linking. *Biomacromolecules* **2016**, *17* (2), 649–660.

(58) Spagnol, C.; Rodrigues, F. H. A.; Neto, A. G. V. C.; Pereira, A. G. B.; Fajardo, A. R.; Radovanovic, E.; Rubira, A. F.; Muniz, E. C. Nanocomposites Based on Poly(acrylamide-Co-Acrylate) and Cellulose Nanowhiskers. *Eur. Polym. J.* **2012**, *48* (3), 454–463.

(59) Spagnol, C.; Rodrigues, F. H. A.; Pereira, A. G. B.; Fajardo, A. R.; Rubira, A. F.; Muniz, E. C. Superabsorbent Hydrogel Nanocomposites Based on Starch-G-Poly(sodium Acrylate) Matrix Filled with Cellulose Nanowhiskers. *Cellulose* **2012**, *19* (4), 1225–1237.

(60) Yang, J.; Han, C.-R. R.; Zhang, X.-M. M.; Xu, F.; Sun, R.-C. C. Cellulose Nanocrystals Mechanical Reinforcement in Composite Hydrogels with Multiple Cross-Links: Correlations between Dissipation Properties and Deformation Mechanisms. *Macromolecules* **2014**, *47* (12), 4077–4086.

(61) Zhou, C.; Wu, Q.; Lei, T.; Negulescu, I. I. Adsorption Kinetic and Equilibrium Studies for Methylene Blue Dye by Partially Hydrolyzed Polyacrylamide/cellulose Nanocrystal Nanocomposite Hydrogels. *Chem. Eng. J.* **2014**, *251*, 17–24.

(62) Yang, J.; Zhao, J.-J. J.; Han, C.-R. R.; Duan, J.-F. F.; Xu, F.; Sun, R.-C. C. Tough Nanocomposite Hydrogels from Cellulose Nanocrystals/poly(acrylamide) Clusters: Influence of the Charge Density, Aspect Ratio and Surface Coating with PEG. *Cellulose* **2014**, *21* (1), 541–551.

(63) Kelly, J. A.; Shukaliak, A. M.; Cheung, C. C. Y.; Shopsowitz, K. E.; Hamad, W. Y.; MacLachlan, M. J. Responsive Photonic Hydrogels Based on Nanocrystalline Cellulose. *Angew. Chem., Int. Ed.* **2013**, *52* (34), 8912–8916.

(64) Yang, X.; Bakaic, E.; Hoare, T.; Cranston, E. D. Injectable Polysaccharide Hydrogels Reinforced with Cellulose Nanocrystals: Morphology, Rheology, Degradation, and Cytotoxicity. *Biomacromolecules* **2013**, *14* (12), 4447–4455.

(65) Yang, J.; Han, C. R.; Duan, J. F.; Xu, F.; Sun, R. C. Mechanical and Viscoelastic Properties of Cellulose Nanocrystals Reinforced Poly(ethylene Glycol) Nanocomposite Hydrogels. *ACS Appl. Mater. Interfaces* **2013**, *5* (8), 3199–3207.

(66) Yang, J.; Zhang, X.-M.; Xu, F. Design of Cellulose Nanocrystals Template-Assisted Composite Hydrogels: Insights from Static to Dynamic Alignment. *Macromolecules* **2015**, *48*, 1231–1239.

(67) Lin, N.; Dufresne, A. Supramolecular Hydrogels from in Situ Host-Guest Inclusion between Chemically Modified Cellulose Nanocrystals and Cyclodextrin. *Biomacromolecules* **2013**, *14* (3), 871–880.

(68) Mohammed, N.; Grishkewich, N.; Berry, R. M.; Tam, K. C. Cellulose Nanocrystals-Alginate Hydrogel Beads as Novel Adsorbents for Organic Dyes in Aqueous Solutions. *Cellulose* **2015**, *22* (6), 3725–3738.

(69) Mohammed, N.; Grishkewich, N.; Waeijen, H. A.; Berry, R. M.; Tam, K. C. Continuous Flow Adsorption of Methylene Blue by Cellulose Nanocrystals-Alginate Hydrogel Beads in Packed Columns. *Carbohydr. Polym.* **2016**, *136*, 1194–1202.

(70) Lin, N.; Gèze, A.; Wouessidjewe, D.; Huang, J.; Dufresne, A. Biocompatible Double-Membrane Hydrogels from Cationic Cellulose Nanocrystals and Anionic Alginate as Complexing Drugs Co-Delivery. *ACS Appl. Mater. Interfaces* **2016**, *8* (11), 6880–6889.

(71) Wang, K.; Nune, K. C.; Misra, R. D. K. The Functional Response of Alginate-Gelatin-Nanocrystalline Cellulose Injectable Hydrogels toward Delivery of Cells and Bioactive Molecules. *Acta Biomater.* **2016**, *36*, 143–151.

(72) Yin, O. S.; Ahmad, I.; Amin, M. C. I. M. Effect of Cellulose Nanocrystals Content and pH on Swelling Behaviour of Gelatin Based Hydrogel. *Sains Malays.* **2015**, *44* (6), 793–799.

(73) Ooi, S. Y.; Ahmad, I.; Amin, M. C. I. M. Cellulose Nanocrystals Extracted from Rice Husks as a Reinforcing Material in Gelatin

Hydrogels for Use in Controlled Drug Delivery Systems. *Ind. Crops Prod.* **2016**, 93, 227–234.

(74) Osorio-Madrado, A.; Eder, M.; Rueggeberg, M.; Pandey, J. K.; Harrington, M. J.; Nishiyama, Y.; Putaux, J.-L.; Rochas, C.; Burgert, I. Reorientation of Cellulose Nanowhiskers in Agarose Hydrogels under Tensile Loading. *Biomacromolecules* **2012**, 13 (3), 850–856.

(75) Le Goff, K. J.; Gaillard, C.; Helbert, W.; Garnier, C.; Aubry, T. Rheological Study of Reinforcement of Agarose Hydrogels by Cellulose Nanowhiskers. *Carbohydr. Polym.* **2015**, 116, 117–123.

(76) Araki, J.; Yamanaka, Y. Anionic and Cationic Nanocomposite Hydrogels Reinforced with Cellulose and Chitin Nanowhiskers: Effect of Electrolyte Concentration on Mechanical Properties and Swelling Behaviors. *Polym. Adv. Technol.* **2014**, 25 (10), 1108–1115.

(77) Gao, X.; Sadasivuni, K. K.; Kim, H.-C.; Min, S.-K.; Kim, J. Designing pH-Responsive and Dielectric Hydrogels from Cellulose Nanocrystals. *J. Chem. Sci.* **2015**, 127 (6), 1119–1125.

(78) McKee, J. R.; Hietala, S.; Seitonen, J.; Laine, J.; Kontturi, E.; Ikkala, O. Thermoresponsive Nanocellulose Hydrogels with Tunable Mechanical Properties. *ACS Macro Lett.* **2014**, 3, 266–270.

(79) Wang, Y.; Chen, L. Impacts of Nanowhisker on Formation Kinetics and Properties of All-Cellulose Composite Gels. *Carbohydr. Polym.* **2011**, 83 (4), 1937–1946.

(80) Dai, Q.; Kadla, J. F. Effect of Nanofillers on Carboxymethyl Cellulose/ Hydroxyethyl Cellulose Hydrogels. *J. Appl. Polym. Sci.* **2009**, 114, 1664–1669.

(81) Wu, Q.; Meng, Y.; Wang, S.; Li, Y.; Fu, S.; Ma, L.; Harper, D. Rheological Behavior of Cellulose Nanocrystal Suspension: Influence of Concentration and Aspect Ratio. *J. Appl. Polym. Sci.* **2014**, 131 (15), 40525–40533.

(82) Sanna, R.; Fortunati, E.; Alzari, V.; Nuvoli, D.; Terenzi, A.; Casula, M. F.; Kenny, J. M.; Mariani, A. Poly(N-Vinylcaprolactam) Nanocomposites Containing Nanocrystalline Cellulose: A Green Approach to Thermoresponsive Hydrogels. *Cellulose* **2013**, 20 (5), 2393–2402.

(83) Yang, J.; Han, C. Mechanically Viscoelastic Properties of Cellulose Nanocrystals Skeleton Reinforced Hierarchical Composite Hydrogels. *ACS Appl. Mater. Interfaces* **2016**, 8 (38), 25621–25630.

(84) Spagnol, C.; Rodrigues, F. H. a; Pereira, A. G. B.; Fajardo, A. R.; Rubira, A. F.; Muniz, E. C. Superabsorbent Hydrogel Composite Made of Cellulose Nanofibrils and Chitosan-Graft-Poly(acrylic Acid). *Carbohydr. Polym.* **2012**, 87 (3), 2038–2045.

(85) Yoshitake, H.; Sugimura, K.; Teramoto, Y.; Nishio, Y. Magnetic Property of Oriented Films of Cellulose Nanocrystal/carrageenan Composites Containing Iron Oxide Nanoparticles: Effect of Anisotropic Aggregation of the Nanoparticles. *Polymer* **2016**, 99, 147–156.

(86) You, J.; Cao, J.; Zhao, Y.; Zhang, L.; Zhou, J.; Chen, Y. Improved Mechanical Properties and Sustained Release Behavior of Cationic Cellulose Nanocrystals Reinforced Cationic Cellulose Injectable Hydrogels. *Biomacromolecules* **2016**, 17 (9), 2839–2848.

(87) Zhou, C.; Wu, Q.; Yue, Y.; Zhang, Q. Application of Rod-Shaped Cellulose Nanocrystals in Polyacrylamide Hydrogels. *J. Colloid Interface Sci.* **2011**, 353 (1), 116–123.

(88) Zhou, C.; Wu, Q.; Zhang, Q. Dynamic Rheology Studies of in Situ Polymerization Process of Polyacrylamide-Cellulose Nanocrystal Composite Hydrogels. *Colloid Polym. Sci.* **2011**, 289 (3), 247–255.

(89) Hebeish, A.; Farag, S.; Sharaf, S.; Shaheen, T. I. Thermal Responsive Hydrogels Based on Semi Interpenetrating Network of poly(NIPAm) and Cellulose Nanowhiskers. *Carbohydr. Polym.* **2014**, 102, 159–166.

(90) Yang, J.; Han, C. R.; Duan, J. F.; Ma, M. G.; Zhang, X. M.; Xu, F.; Sun, R. C. Synthesis and Characterization of Mechanically Flexible and Tough Cellulose Nanocrystals-Polyacrylamide Nanocomposite Hydrogels. *Cellulose* **2013**, 20 (1), 227–237.

(91) Yang, J.; Zhao, J.-J.; Zhang, X.-M. Modification of Cellulose Nanocrystal-Reinforced Composite Hydrogels: Effects of Co-Cross-linked and Drying Treatment. *Cellulose* **2014**, 21 (5), 3487–3496.

(92) Yang, J.; Han, C.-R.; Duan, J.-F.; Ma, M.-G.; Zhang, X.-M.; Xu, F.; Sun, R.-C.; Xie, X.-M. Studies on the Properties and Formation

Mechanism of Flexible Nanocomposite Hydrogels from Cellulose Nanocrystals and Poly(acrylic Acid). *J. Mater. Chem.* **2012**, 22 (42), 22467–22480.

(93) Yang, J.; Zhao, J.-J.; Xu, F.; Sun, R.-C. Revealing Strong Nanocomposite Hydrogels Reinforced by Cellulose Nanocrystals: Insight into Morphologies and Interactions. *ACS Appl. Mater. Interfaces* **2013**, 5 (24), 12960–12967.

(94) Yang, J.; Han, C.; Xu, F.; Sun, R. Simple Approach to Reinforce Hydrogels with Cellulose Nanocrystals. *Nanoscale* **2014**, 6 (11), 5934–5943.

(95) Atifi, S.; Su, S.; Hamad, W. Y. Mechanically Tunable Nanocomposite Hydrogels Based on Functionalized Cellulose Nanocrystals. *Nord. Pulp Pap. Res. J.* **2014**, 29 (1), 95–104.

(96) Yang, D.; Peng, X.; Zhong, L.; Cao, X.; Chen, W.; Wang, S.; Liu, C.; Sun, R. Fabrication of a Highly Elastic Nanocomposite Hydrogel by Surface Modification of Cellulose Nanocrystals. *RSC Adv.* **2015**, 5 (18), 13878–13885.

(97) Chau, M.; De France, K. J.; Kopera, B.; Machado, V. R.; Rosenfeldt, S.; Reyes, L.; Chan, K. J. W.; Förster, S.; Cranston, E. D.; Hoare, T.; et al. Composite Hydrogels with Tunable Anisotropic Morphologies and Mechanical Properties. *Chem. Mater.* **2016**, 28 (10), 3406–3415.

(98) Dugan, J. M.; Gough, J. E.; Eichhorn, S. J. Directing the Morphology and Differentiation of Skeletal Muscle Cells Using Oriented Cellulose Nanowhiskers. *Biomacromolecules* **2010**, 11 (9), 2498–2504.

(99) Domingues, R. M. A.; Silva, M.; Gershovich, P.; Betta, S.; Babo, P.; Caridade, S. G.; Mano, J. F.; Motta, A.; Reis, R. L.; Gomes, M. E. Development of Injectable Hyaluronic Acid/Cellulose Nanocrystals Bionanocomposite Hydrogels for Tissue Engineering Applications. *Bioconjugate Chem.* **2015**, 26 (8), 1571–1581.

(100) Köhnke, T.; Elder, T.; Theliander, H.; Ragauskas, A. J. Ice Templated and Cross-Linked Xylan/nanocrystalline Cellulose Hydrogels. *Carbohydr. Polym.* **2014**, 100, 24–30.

(101) Dash, R.; Foston, M.; Ragauskas, A. J. Improving the Mechanical and Thermal Properties of Gelatin Hydrogels Cross-Linked by Cellulose Nanowhiskers. *Carbohydr. Polym.* **2013**, 91 (2), 638–645.

(102) García-Astrain, C.; González, K.; Gurrea, T.; Guaresti, O.; Algar, I.; Eceiza, A.; Gabilondo, N. Maleimide-Grafted Cellulose Nanocrystals As Cross-Linkers for Bionanocomposite Hydrogels. *Carbohydr. Polym.* **2016**, 149, 94–101.

(103) Li, W.; Lan, Y.; Guo, R.; Zhang, Y.; Xue, W.; Zhang, Y. In Vitro and in Vivo Evaluation of a Novel Collagen/cellulose Nanocrystals Scaffold for Achieving the Sustained Release of Basic Fibroblast Growth Factor. *J. Biomater. Appl.* **2015**, 29 (6), 882–893.

(104) Mauricio, M. R.; Da Costa, P. G.; Haraguchi, S. K.; Guilherme, M. R.; Muniz, E. C.; Rubira, A. F. Synthesis of a Microhydrogel Composite from Cellulose Nanowhiskers and Starch for Drug Delivery. *Carbohydr. Polym.* **2015**, 115, 715–722.

(105) Mihranyan, A. Viscoelastic Properties of Cross-Linked Polyvinyl Alcohol and Surface-Oxidized Cellulose Whisker Hydrogels. *Cellulose* **2013**, 20 (3), 1369–1376.

(106) Anirudhan, T. S.; Rejeena, S. R. Poly(acrylic Acid- Co -Acrylamide- Co -2-Acrylamido-2-Methyl-1-Propanesulfonic Acid)-Grafted Nanocellulose/poly(vinyl Alcohol) Composite for the in Vitro Gastrointestinal Release of Amoxicillin. *J. Appl. Polym. Sci.* **2014**, 131 (17), 40699–40711.

(107) Goetz, L.; Foston, M.; Mathew, A. P.; Oksman, K.; Ragauskas, A. J. Poly(methyl Vinyl Ether-Co-Maleic Acid)-Polyethylene Glycol Nanocomposites Cross-Linked in Situ with Cellulose Nanowhiskers. *Biomacromolecules* **2010**, 11 (10), 2660–2666.

(108) Fall, A. B.; Lindström, S. B.; Sundman, O.; Ödberg, L.; Wågberg, L. Colloidal Stability of Aqueous Nanofibrillated Cellulose Dispersions. *Langmuir* **2011**, 27 (18), 11332–11338.

(109) Saito, T.; Uematsu, T.; Kimura, S.; Enomae, T.; Isogai, A. Self-Aligned Integration of Native Cellulose Nanofibrils towards Producing Diverse Bulk Materials. *Soft Matter* **2011**, 7 (19), 8804–8809.

- (110) Fukuzumi, H.; Tanaka, R.; Saito, T.; Isogai, A. Dispersion Stability and Aggregation Behavior of TEMPO-Oxidized Cellulose Nanofibrils in Water as a Function of Salt Addition. *Cellulose* **2014**, *21* (3), 1553–1559.
- (111) Fall, A. B.; Lindström, S. B.; Sprakel, J.; Wågberg, L. A Physical Cross-Linking Process of Cellulose Nanofibril Gels with Shear-Controlled Fibril Orientation. *Soft Matter* **2013**, *9* (6), 1852–1863.
- (112) Håkansson, K. M. O.; Fall, A. B.; Lundell, F.; Yu, S.; Krywka, C.; Roth, S. V.; Santoro, G.; Kvick, M.; Prahl Wittberg, L.; Wågberg, L.; et al. Hydrodynamic Alignment and Assembly of Nanofibrils Resulting in Strong Cellulose Filaments. *Nat. Commun.* **2014**, *5*, 4018–4027.
- (113) Lundahl, M. J.; Cunha, A. G.; Rojo, E.; Papageorgiou, A. C.; Rautkari, L.; Arboleda, J. C.; Rojas, O. J. Strength and Water Interactions of Cellulose I Filaments Wet-Spun from Cellulose Nanofibril Hydrogels. *Sci. Rep.* **2016**, *6*, 30695–30707.
- (114) Abe, K.; Yano, H. Formation of Hydrogels from Cellulose Nanofibers. *Carbohydr. Polym.* **2011**, *85* (4), 733–737.
- (115) Abe, K.; Yano, H. Cellulose Nanofiber-Based Hydrogels with High Mechanical Strength. *Cellulose* **2012**, *19* (6), 1907–1912.
- (116) Bhattacharya, M.; Malinen, M. M.; Lauren, P.; Lou, Y. R.; Kuisma, S. W.; Kanninen, L.; Lille, M.; Corlu, A.; Guguen-Guillouzo, C.; Ikkala, O.; et al. Nanofibrillar Cellulose Hydrogel Promotes Three-Dimensional Liver Cell Culture. *J. Controlled Release* **2012**, *164* (3), 291–298.
- (117) Dong, H.; Snyder, J. F.; Williams, K. S.; Andzelm, J. W. Cation-Induced Hydrogels of Cellulose Nanofibrils with Tunable Moduli. *Biomacromolecules* **2013**, *14* (9), 3338–3345.
- (118) Masruchin, N.; Park, B.-D.; Causin, V.; Um, I. C. Characteristics of TEMPO-Oxidized Cellulose Fibril-Based Hydrogels Induced by Cationic Ions and Their Properties. *Cellulose* **2015**, *22* (3), 1993–2010.
- (119) Dong, H.; Snyder, J. F.; Tran, D. T.; Leadore, J. L. Hydrogel, Aerogel and Film of Cellulose Nanofibrils Functionalized with Silver Nanoparticles. *Carbohydr. Polym.* **2013**, *95* (2), 760–767.
- (120) Zander, N. E.; Dong, H.; Steele, J.; Grant, J. T. Metal Cation Cross-Linked Nanocellulose Hydrogels as Tissue Engineering Substrates. *ACS Appl. Mater. Interfaces* **2014**, *6* (21), 18502–18510.
- (121) Ma, H.; Hsiao, B. S.; Chu, B. Ultrafine Cellulose Nanofibers as Efficient Adsorbents for Removal of UO₂ in Water. *ACS Macro Lett.* **2012**, *1*, 213–216.
- (122) Torres-Rendon, J. G.; Femmer, T.; De Laporte, L.; Tigges, T.; Rahimi, K.; Gremse, F.; Zafarnia, S.; Lederle, W.; Ifuku, S.; Wessling, M.; et al. Bioactive Gyroid Scaffolds Formed by Sacrificial Templating of Nanocellulose and Nanochitin Hydrogels as Instructive Platforms for Biomimetic Tissue Engineering. *Adv. Mater.* **2015**, *27*, 2989–2995.
- (123) Lauren, P.; Lou, Y. R.; Raki, M.; Urtti, A.; Bergstrom, K.; Yliperttula, M. Technetium-99m-Labeled Nanofibrillar Cellulose Hydrogel for in Vivo Drug Release. *Eur. J. Pharm. Sci.* **2014**, *65*, 79–88.
- (124) Borges, A. C.; Eyholzer, C.; Duc, F.; Bourban, P.-E.; Tingaut, P.; Zimmermann, T.; Pioletti, D. P.; Manson, J.-A. E. Nanofibrillated Cellulose Composite Hydrogel for the Replacement of the Nucleus Pulposus. *Acta Biomater.* **2011**, *7* (9), 3412–3421.
- (125) Eyholzer, C.; Borges De Couraça, A.; Duc, F.; Bourban, P. E.; Tingaut, P.; Zimmermann, T.; Manson, J. A. E.; Oksman, K. Biocomposite Hydrogels with Carboxymethylated, Nanofibrillated Cellulose Powder for Replacement of the Nucleus Pulposus. *Biomacromolecules* **2011**, *12* (5), 1419–1427.
- (126) Mathew, A. P.; Oksman, K.; Pierron, D.; Harmad, M. F. Crosslinked Fibrous Composites Based on Cellulose Nanofibers and Collagen with in Situ pH Induced Fibrillation. *Cellulose* **2012**, *19*, 139–150.
- (127) Mathew, A. P.; Oksman, K.; Pierron, D.; Harmand, M. F. Biocompatible Fibrous Networks of Cellulose Nanofibres and Collagen Crosslinked Using Genipin: Potential as Artificial Ligament/Tendons. *Macromol. Biosci.* **2013**, *13*, 289–298.
- (128) Naseri, N.; Poirier, J.-M.; Girandon, L.; Fröhlich, M.; Oksman, K.; Mathew, A. P. 3-Dimensional Porous Nanocomposite Scaffolds Based on Cellulose Nanofibers for Cartilage Tissue Engineering: Tailoring of Porosity and Mechanical Performance. *RSC Adv.* **2016**, *6* (8), 5999–6007.
- (129) Wang, B.; Benitez, A. J.; Lossada, F.; Merindol, R.; Walther, A. Bioinspired Mechanical Gradients in Cellulose Nanofibril/Polymer Nanopapers. *Angew. Chem., Int. Ed.* **2016**, *55* (20), 5966–5970.
- (130) Yang, J.; Zhang, X.; Ma, M.; Xu, F. Modulation of Assembly and Dynamics in Colloidal Hydrogels via Ionic Bridge from Cellulose Nanofibrils and Poly(ethylene Glycol). *ACS Macro Lett.* **2015**, *4*, 829–833.
- (131) Wei, J.; Chen, Y.; Liu, H.; Du, C.; Yu, H.; Zhou, Z. Thermo-Responsive and Compression Properties of TEMPO-Oxidized Cellulose Nanofiber-Modified PNIPAm Hydrogels. *Carbohydr. Polym.* **2016**, *147*, 201–207.
- (132) Syverud, K.; Kirsebom, H.; Hajizadeh, S.; Chinga-Carrasco, G. Cross-Linking Cellulose Nanofibrils for Potential Elastic Cryo-Structured Gels. *Nanoscale Res. Lett.* **2011**, *6* (1), 626–631.
- (133) Zhou, Y.; Fu, S.; Zhang, L.; Zhan, H.; Levit, M. V. Use of Carboxylated Cellulose Nanofibrils-Filled Magnetic Chitosan Hydrogel Beads as Adsorbents for Pb(II). *Carbohydr. Polym.* **2014**, *101*, 75–82.
- (134) Zhou, Y.; Fu, S.; Zhang, L.; Zhan, H. Superabsorbent Nanocomposite Hydrogels Made of Carboxylated Cellulose Nanofibrils and CMC-G-p(AA-Co-AM). *Carbohydr. Polym.* **2013**, *97* (2), 429–435.
- (135) Yue, Y.; Han, J.; Han, G.; French, A. D.; Qi, Y.; Wu, Q. Cellulose Nanofibers Reinforced Sodium Alginate-Polyvinyl Alcohol Hydrogels: Core-Shell Structure Formation and Property Characterization. *Carbohydr. Polym.* **2016**, *147*, 155–164.
- (136) Chinga-Carrasco, G.; Syverud, K. Pretreatment-Dependent Surface Chemistry of Wood Nanocellulose for pH-Sensitive Hydrogels. *J. Biomater. Appl.* **2014**, *29* (7491), 423–432.
- (137) Olszewska, A.; Eronen, P.; Johansson, L. S.; Malho, J. M.; Ankerfors, M.; Lindström, T.; Ruokolainen, J.; Laine, J.; Österberg, M. The Behaviour of Cationic Nanofibrillar Cellulose in Aqueous Media. *Cellulose* **2011**, *18* (5), 1213–1226.
- (138) Fujisawa, S.; Okita, Y.; Fukuzumi, H.; Saito, T.; Isogai, A. Preparation and Characterization of TEMPO-Oxidized Cellulose Nanofibril Films with Free Carboxyl Groups. *Carbohydr. Polym.* **2011**, *84* (1), 579–583.
- (139) Malho, J. M.; Laaksonen, P.; Walther, A.; Ikkala, O.; Linder, M. B. Facile Method for Stiff, Tough, and Strong Nanocomposites by Direct Exfoliation of Multilayered Graphene into Native Nanocellulose Matrix. *Biomacromolecules* **2012**, *13* (4), 1093–1099.
- (140) Rees, A.; Powell, L. C.; Chinga-Carrasco, G.; Gethin, D. T.; Syverud, K.; Hill, K. E.; Thomas, D. W. 3D Bioprinting of Carboxymethylated-Periodate Oxidized Nanocellulose Constructs for Wound Dressing Applications. *BioMed Res. Int.* **2015**, *2015*, 1–7.
- (141) Spoljaric, S.; Salminen, A.; Luong, N. D.; Seppälä, J. Stable, Self-Healing Hydrogels from Nanofibrillated Cellulose, Poly(vinyl Alcohol) and Borax via Reversible Crosslinking. *Eur. Polym. J.* **2014**, *56* (1), 105–117.
- (142) Yang, J.; Xu, F.; Han, C.-R. Metal Ion Mediated Cellulose Nanofibrils Transient Network in Covalently Cross-Linked Hydrogels: Mechanistic Insight into Morphology and Dynamics. *Biomacromolecules* **2017**, *18*, 1019.
- (143) Liu, J.; Chinga-Carrasco, G.; Cheng, F.; Xu, W.; Willför, S.; Syverud, K.; Xu, C. Hemicellulose-Reinforced Nanocellulose Hydrogels for Wound Healing Application. *Cellulose* **2016**, *23*, 3129–3144.
- (144) Müller, M.; Ozturk, E.; Arlov, O.; Gatenholm, P.; Zenobi-Wong, M. Alginate Sulfate-Nanocellulose Bioinks for Cartilage Bioprinting Applications. *Ann. Biomed. Eng.* **2017**, *45* (1), 210–223.
- (145) Alexandrescu, L.; Syverud, K.; Gatti, A.; Chinga-Carrasco, G. Cytotoxicity Tests of Cellulose Nanofibril-Based Structures. *Cellulose* **2013**, *20* (4), 1765–1775.
- (146) Dash, R.; Cateto, C. A.; Ragauskas, A. J. Synthesis of a Co-Cross-Linked Nanocomposite Hydrogels from Poly(methyl Vinyl Ether-Co-Maleic Acid)-Polyethylene Glycol and Nanofibrillated Cellulose. *Cellulose* **2014**, *21* (1), 529–534.

- (147) Syverud, K.; Pettersen, S. R.; Draget, K.; Chinga-Carrasco, G. Controlling the Elastic Modulus of Cellulose Nanofibril Hydrogels-Scaffolds with Potential in Tissue Engineering. *Cellulose* **2015**, *22* (1), 473–481.
- (148) Yang, X.; Cranston, E. D. Chemically Cross-Linked Cellulose Nanocrystal Aerogels with Shape Recovery and Superabsorbent Properties. *Chem. Mater.* **2014**, *26* (20), 6016–6025.
- (149) Heath, L.; Thielemans, W. Cellulose Nanowhisker Aerogels. *Green Chem.* **2010**, *12* (8), 1448–1453.
- (150) Kobayashi, Y.; Saito, T.; Isogai, A. Aerogels with 3D Ordered Nanofiber Skeletons of Liquid-Crystalline Nanocellulose Derivatives as Tough and Transparent Insulators. *Angew. Chem.* **2014**, *126*, 10562–10565.
- (151) Yang, X.; Shi, K.; Zhitomirsky, I.; Cranston, E. D. Cellulose Nanocrystal Aerogels as Universal 3D Lightweight Substrates for Supercapacitor Materials. *Adv. Mater.* **2015**, *27* (40), 6104–6109.
- (152) Shi, K.; Yang, X.; Cranston, E. D.; Zhitomirsky, I. Efficient Lightweight Supercapacitor with Compression Stability. *Adv. Funct. Mater.* **2016**, *26* (35), 6437–6445.
- (153) Zhu, H.; Yang, X.; Cranston, E. D.; Zhu, S. Flexible and Porous Nanocellulose Aerogels with High Loadings of Metal-Organic Framework Particles for Separations Applications. *Adv. Mater.* **2016**, *28* (35), 7652–7657.
- (154) Fumagalli, M.; Sanchez, F.; Boisseau, S. M.; Heux, L. Gas-Phase Esterification of Cellulose Nanocrystal Aerogels for Colloidal Dispersion in Apolar Solvents. *Soft Matter* **2013**, *9* (47), 11309–11317.
- (155) Ahmadi, M.; Madadlou, A.; Saboury, A. A. Whey Protein Aerogel as Blended with Cellulose Crystalline Particles or Loaded with Fish Oil. *Food Chem.* **2016**, *196*, 1016–1022.
- (156) Gawryla, M. D.; van den Berg, O.; Weder, C.; Schiraldi, D. A. Clay Aerogel/cellulose Whisker Nanocomposites: A Nanoscale Wattle and Daub. *J. Mater. Chem.* **2009**, *19* (15), 2118–2124.
- (157) Huang, P.; Fan, M. Development of Fracture Free Clay-Based Aerogel: Formulation and Architectural Mechanisms. *Composites, Part B* **2016**, *91*, 169–175.
- (158) Shamskar, K. R.; Heidari, H.; Rashidi, A. Preparation and Evaluation of Nanocrystalline Cellulose Aerogels from Raw Cotton and Cotton Stalk. *Ind. Crops Prod.* **2016**, *93*, 203–211.
- (159) Buesch, C.; Smith, S. W.; Eschbach, P.; Conley, J. F.; Simonsen, J. The Microstructure of Cellulose Nanocrystal Aerogels as Revealed by Transmission Electron Microscope Tomography. *Biomacromolecules* **2016**, *17* (9), 2956–2962.
- (160) Wang, X.; Zhang, Y.; Jiang, H.; Song, Y.; Zhou, Z.; Zhao, H. Fabrication and Characterization of Nano-Cellulose Aerogels via Supercritical CO₂ Drying Technology. *Mater. Lett.* **2016**, *183*, 179–182.
- (161) Abraham, E.; Weber, D. E.; Sharon, S.; Lapidot, S.; Shoseyov, O. Multifunctional Cellulosic Scaffolds from Modified Cellulose Nanocrystals. *ACS Appl. Mater. Interfaces* **2017**, *9* (3), 2010–2015.
- (162) Kumar, A.; Lee, Y.; Kim, D.; Rao, K. M.; Kim, J.; Park, S.; Haider, A.; Lee, D. H.; Han, S. S. Effect of Crosslinking Functionality on Microstructure, Mechanical Properties, and in Vitro Cytocompatibility of Cellulose Nanocrystals Reinforced Poly (Vinyl Alcohol)/sodium Alginate Hybrid Scaffolds. *Int. J. Biol. Macromol.* **2017**, *95*, 962–973.
- (163) Toivonen, M. S.; Kaskela, A.; Rojas, O. J.; Kauppinen, E. I.; Ikkala, O. Ambient-Dried Cellulose Nanofibril Aerogel Membranes with High Tensile Strength and Their Use for Aerosol Collection and Templates for Transparent, Flexible Devices. *Adv. Funct. Mater.* **2015**, *25* (42), 6618–6626.
- (164) Chen, W.; Li, Q.; Wang, Y.; Yi, X.; Zeng, J.; Yu, H.; Liu, Y.; Li, J. Comparative Study of Aerogels Obtained from Differently Prepared Nanocellulose Fibers. *ChemSusChem* **2014**, *7* (1), 154–161.
- (165) Chen, W.; Yu, H.; Li, Q.; Liu, Y.; Li, J. Ultralight and Highly Flexible Aerogels with Long Cellulose I Nanofibers. *Soft Matter* **2011**, *7* (21), 10360–10368.
- (166) Munier, P.; Gordeyeva, K.; Bergström, L.; Fall, A. B. Directional Freezing of Nanocellulose Dispersions Aligns the Rod-like Particles and Produces Low-Density and Robust Particle Networks. *Biomacromolecules* **2016**, *17*, 1875–1881.
- (167) Jiang, F.; Hsieh, Y.-L. Amphiphilic Superabsorbent Cellulose Nanofibril Aerogels. *J. Mater. Chem. A* **2014**, *2* (18), 6337–6342.
- (168) Jiang, F.; Hsieh, Y.-L. Super Water Absorbing and Shape Memory Nanocellulose Aerogels from TEMPO-Oxidized Cellulose Nanofibrils via Cyclic Freezing–thawing. *J. Mater. Chem. A* **2014**, *2* (2), 350–359.
- (169) Kettunen, M.; Silvennoinen, R. J.; Houbenov, N.; Nykänen, A.; Ruokolainen, J.; Sainio, J.; Pore, V.; Kemell, M.; Ankerfors, M.; Lindström, T.; et al. Photoswitchable Superabsorbency Based on Nanocellulose Aerogels. *Adv. Funct. Mater.* **2011**, *21* (3), 510–517.
- (170) Aulin, C.; Netrval, J.; Wågberg, L.; Lindström, T. Aerogels from Nanofibrillated Cellulose with Tunable Oleophobicity. *Soft Matter* **2010**, *6* (14), 3298–3305.
- (171) Cervin, N. T.; Aulin, C.; Larsson, P. T.; Wågberg, L. Ultra Porous Nanocellulose Aerogels as Separation Medium for Mixtures of Oil/water Liquids. *Cellulose* **2012**, *19* (2), 401–410.
- (172) Jin, H.; Kettunen, M.; Laiho, A.; Pynnönen, H.; Paltakari, J.; Marmur, A.; Ikkala, O.; Ras, R. H. A. Superhydrophobic and Superoleophobic Nanocellulose Aerogel Membranes as Bioinspired Cargo Carriers on Water and Oil. *Langmuir* **2011**, *27* (5), 1930–1934.
- (173) Korhonen, J. T.; Kettunen, M.; Ras, R. H. a; Ikkala, O. Hydrophobic Nanocellulose Aerogels as Floating, Sustainable, Reusable, and Recyclable Oil Absorbents. *ACS Appl. Mater. Interfaces* **2011**, *3* (6), 1813–1816.
- (174) Xiao, S.; Gao, R.; Lu, Y.; Li, J.; Sun, Q. Fabrication and Characterization of Nanofibrillated Cellulose and Its Aerogels from Natural Pine Needles. *Carbohydr. Polym.* **2015**, *119*, 202–209.
- (175) Miettunen, K.; Vapaavuori, J.; Tiuhonen, A.; Poskela, A.; Lahtinen, P.; Halme, J.; Lund, P. Nanocellulose Aerogel Membranes for Optimal Electrolyte Filling in Dye Solar Cells. *Nano Energy* **2014**, *8*, 95–102.
- (176) Zu, G.; Shen, J.; Zou, L.; Wang, F.; Wang, X.; Zhang, Y.; Yao, X. Nanocellulose-Derived Highly Porous Carbon Aerogels for Supercapacitors. *Carbon* **2016**, *99*, 203–211.
- (177) Pääkkö, M.; Vapaavuori, J.; Silvennoinen, R.; Kosonen, H.; Ankerfors, M.; Lindström, T.; Berglund, L. a; Ikkala, O. Long and Entangled Native Cellulose I Nanofibers Allow Flexible Aerogels and Hierarchically Porous Templates for Functionalities. *Soft Matter* **2008**, *4* (12), 2492–2499.
- (178) Korhonen, J. T.; Hiekkataipale, P.; Malm, J.; Karppinen, M.; Ikkala, O.; Ras, R. H. A. Inorganic Hollow Nanotube Aerogels by Atomic Layer Deposition onto Native Nanocellulose Templates. *ACS Nano* **2011**, *5* (3), 1967–1974.
- (179) Nyström, G.; Marais, A.; Karabulut, E.; Wågberg, L.; Cui, Y.; Hamed, M. M. Self-Assembled Three-Dimensional and Compressible Interdigitated Thin-Film Supercapacitors and Batteries. *Nat. Commun.* **2015**, *6*, 7259–7266.
- (180) Hamed, M.; Karabulut, E.; Marais, A.; Herland, A.; Nyström, G.; Wågberg, L. Nanocellulose Aerogels Functionalized by Rapid Layer-by-Layer Assembly for High Charge Storage and Beyond. *Angew. Chem., Int. Ed.* **2013**, *52* (46), 12038–12042.
- (181) Carlsson, D. O.; Nyström, G.; Zhou, Q.; Berglund, L. A.; Nyholm, L.; Strømme, M. Electroactive Nanofibrillated Cellulose Aerogel Composites with Tunable Structural and Electrochemical Properties. *J. Mater. Chem.* **2012**, *22* (36), 19014–19024.
- (182) Nyström, G.; Mihranyan, A.; Razaq, A.; Lindström, T.; Nyholm, L.; Strømme, M. A Nanocellulose Polypyrrole Composite Based on Microfibrillated Cellulose from Wood. *J. Phys. Chem. B* **2010**, *114* (12), 4178–4182.
- (183) Gao, K.; Shao, Z.; Li, J.; Wang, X.; Peng, X.; Wang, W.; Wang, F. Cellulose Nanofiber–graphene All Solid-State Flexible Supercapacitors. *J. Mater. Chem. A* **2013**, *1* (1), 63–67.
- (184) Gao, K.; Shao, Z.; Wang, X.; Zhang, Y.; Wang, W.; Wang, F. Cellulose Nanofibers/multi-Walled Carbon Nanotube Nanohybrid Aerogel for All-Solid-State Flexible Supercapacitors. *RSC Adv.* **2013**, *3* (35), 15058–15064.

- (185) Wang, M.; Anoshkin, I. V.; Nasibulin, A. G.; Korhonen, J. T.; Seitsonen, J.; Pere, J.; Kauppinen, E. I.; Ras, R. H. A.; Ikkala, O. Modifying Native Nanocellulose Aerogels with Carbon Nanotubes for Mechanoresponsive Conductivity and Pressure Sensing. *Adv. Mater.* **2013**, *25* (17), 2428–2432.
- (186) Zhai, T.; Zheng, Q.; Cai, Z.; Turng, L.; Xia, H.; Gong, S. Poly(vinyl alcohol)/Cellulose Nanofibril Hybrid Aerogels with an Aligned Microtubular Porous Structure and Their Composites with Polydimethylsiloxane. *ACS Appl. Mater. Interfaces* **2015**, *7*, 7436–7444.
- (187) Melone, L.; Altomare, L.; Alfieri, I.; Lorenzi, A.; De Nardo, L.; Punta, C. Ceramic Aerogels from TEMPO-Oxidized Cellulose Nanofibre Templates: Synthesis, Characterization, and Photocatalytic Properties. *J. Photochem. Photobiol., A* **2013**, *261*, 53–60.
- (188) Arboleda, J. C.; Hughes, M.; Lucia, L. A.; Laine, J.; Ekman, K.; Rojas, O. J. Soy Protein–nanocellulose Composite Aerogels. *Cellulose* **2013**, *20* (5), 2417–2426.
- (189) Zhang, X.; Wang, Y.; Zhao, J.; Xiao, M.; Zhang, W.; Lu, C. Mechanically Strong and Thermally Responsive Cellulose Nanofibers/Poly (N-Isopropylacrylamide) Hybrid Aerogels. *ACS Sustainable Chem. Eng.* **2016**, *4* (8), 4321–4327.
- (190) Zhai, T.; Zheng, Q.; Cai, Z.; Xia, H.; Gong, S. Synthesis of Polyvinyl Alcohol/cellulose Nanofibril Hybrid Aerogel Microspheres and Their Use as Oil/solvent Superabsorbents. *Carbohydr. Polym.* **2016**, *148*, 300–308.
- (191) Jiang, F.; Hsieh, Y.-L. Cellulose Nanofibril Aerogels: Synergistic Improvement of Hydrophobicity, Strength, and Thermal Stability via Cross-Linking with Diisocyanate. *ACS Appl. Mater. Interfaces* **2017**, *9* (3), 2825–2834.
- (192) Henry, A.; Plumejeau, S.; Heux, L.; Louvain, N.; Monconduit, L.; Stievenot, L.; Boury, B. Conversion of Nanocellulose Aerogel into TiO₂ and TiO₂@C Nano-Thorns by Direct Anhydrous Mineralization with TiCl₄. Evaluation of Electrochemical Properties in Li Batteries. *ACS Appl. Mater. Interfaces* **2015**, *7* (27), 14584–14592.
- (193) Zhang, F.; Ren, H.; Dou, J.; Tong, G.; Deng, Y. Cellulose Nanofibril Based-Aerogel Microreactors: A High Efficiency and Easy Recoverable W/O/W Membrane Separation System. *Sci. Rep.* **2017**, *7*, 40096–40102.
- (194) Gebald, C.; Wurzbacher, J. A.; Tingaut, P.; Zimmermann, T.; Steinfeld, A. Amine-Based Nanofibrillated Cellulose as Adsorbent for CO₂ Capture from Air. *Environ. Sci. Technol.* **2011**, *45* (20), 9101–9108.
- (195) Hayase, G.; Kanamori, K.; Abe, K.; Yano, H.; Maeno, A.; Kaji, H.; Nakanishi, K. Polymethylsilsesquioxane-Cellulose Nanofiber Biocomposite Aerogels with High Thermal Insulation, Bendability, and Superhydrophobicity. *ACS Appl. Mater. Interfaces* **2014**, *6* (12), 9466–9471.
- (196) Donius, A. E.; Liu, A.; Berglund, L. A.; Wegst, U. G. K. Superior Mechanical Performance of Highly Porous, Anisotropic Nanocellulose–montmorillonite Aerogels Prepared by Freeze Casting. *J. Mech. Behav. Biomed. Mater.* **2014**, *37*, 88–99.
- (197) Javadi, A.; Zheng, Q.; Payen, F.; Javadi, A.; Altin, Y.; Cai, Z.; Sabo, R.; Gong, S. Polyvinyl Alcohol-Cellulose Nanofibrils-Graphene Oxide Hybrid Organic Aerogels. *ACS Appl. Mater. Interfaces* **2013**, *5* (13), 5969–5975.
- (198) Jimenez-Saelices, C.; Seantier, B.; Cathala, B.; Grohens, Y. Spray Freeze-Dried Nanofibrillated Cellulose Aerogels with Thermal Superinsulating Properties. *Carbohydr. Polym.* **2017**, *157*, 105–113.
- (199) Silva, T. C. F.; Habibi, Y.; Colodette, J. L.; Elder, T.; Lucia, L. A. A Fundamental Investigation of the Microarchitecture and Mechanical Properties of Tempo-Oxidized Nanofibrillated Cellulose (NFC)-Based Aerogels. *Cellulose* **2012**, *19* (6), 1945–1956.
- (200) Lu, T.; Li, Q.; Chen, W.; Yu, H. Composite Aerogels Based on Dialdehyde Nanocellulose and Collagen for Potential Applications as Wound Dressing and Tissue Engineering Scaffold. *Compos. Sci. Technol.* **2014**, *94*, 132–138.
- (201) Sehaqui, H.; Zhou, Q.; Berglund, L. A. High-Porosity Aerogels of High Specific Surface Area Prepared from Nanofibrillated Cellulose (NFC). *Compos. Sci. Technol.* **2011**, *71* (13), 1593–1599.
- (202) Valo, H.; Arola, S.; Laaksonen, P.; Torkkeli, M.; Peltonen, L.; Linder, M. B.; Serimaa, R.; Kuga, S.; Hirvonen, J.; Laaksonen, T. Drug Release from Nanoparticles Embedded in Four Different Nanofibrillar Cellulose Aerogels. *Eur. J. Pharm. Sci.* **2013**, *50* (1), 69–77.
- (203) Lin, J.; Yu, L.; Tian, F.; Zhao, N.; Li, X.; Bian, F.; Wang, J. Cellulose Nanofibrils Aerogels Generated from Jute Fibers. *Carbohydr. Polym.* **2014**, *109*, 35–43.
- (204) Seantier, B.; Bendahou, D.; Bendahou, A.; Grohens, Y.; Kaddami, H. Multi-Scale Cellulose Based New Bio-Aerogel Composites with Thermal Super-Insulating and Tunable Mechanical Properties. *Carbohydr. Polym.* **2016**, *138*, 335–348.
- (205) Zhou, S.; Liu, P.; Wang, M.; Zhao, H.; Yang, J.; Xu, F. Sustainable, Reusable and Superhydrophobic Aerogels from Microfibrillated Cellulose for Highly Effective Oil/Water Separation. *ACS Sustainable Chem. Eng.* **2016**, *4*, 6409–6416.
- (206) Zhang, W.; Zhang, Y.; Lu, C.; Deng, Y. Aerogels from Crosslinked Cellulose Nano/micro-Fibrils and Their Fast Shape Recovery Property in Water. *J. Mater. Chem.* **2012**, *22* (23), 11642–11650.
- (207) Zhang, Z.; Sèbe, G.; Rentsch, D.; Zimmermann, T.; Tingaut, P. Ultralightweight and Flexible Silylated Nanocellulose Sponges for the Selective Removal of Oil from Water. *Chem. Mater.* **2014**, *26* (8), 2659–2668.
- (208) Meng, Y.; Wang, X.; Wu, Z.; Wang, S.; Young, T. M. Optimization of Cellulose Nanofibrils Carbon Aerogel Fabrication Using Response Surface Methodology. *Eur. Polym. J.* **2015**, *73*, 137–148.
- (209) Sun, B.; Hou, Q.; Liu, Z.; Ni, Y. Sodium Periodate Oxidation of Cellulose Nanocrystal and Its Application as a Paper Wet Strength Additive. *Cellulose* **2015**, *22*, 1135.
- (210) Temelli, F.; Seifried, B. *Supercritical Fluid Treatment of High Molecular Weight Biopolymers*. U.S. Patent US2013/0101849A1, 2013.
- (211) Peng, Y.; Gardner, D. J.; Han, Y.; Kiziltas, A.; Cai, Z.; Tshabalala, M. A. Influence of Drying Method on the Material Properties of Nanocellulose I: Thermostability and Crystallinity. *Cellulose* **2013**, *20* (5), 2379–2392.
- (212) Seifried, B. PGX Technology; <http://ceapro.com/pgx-technology/>.